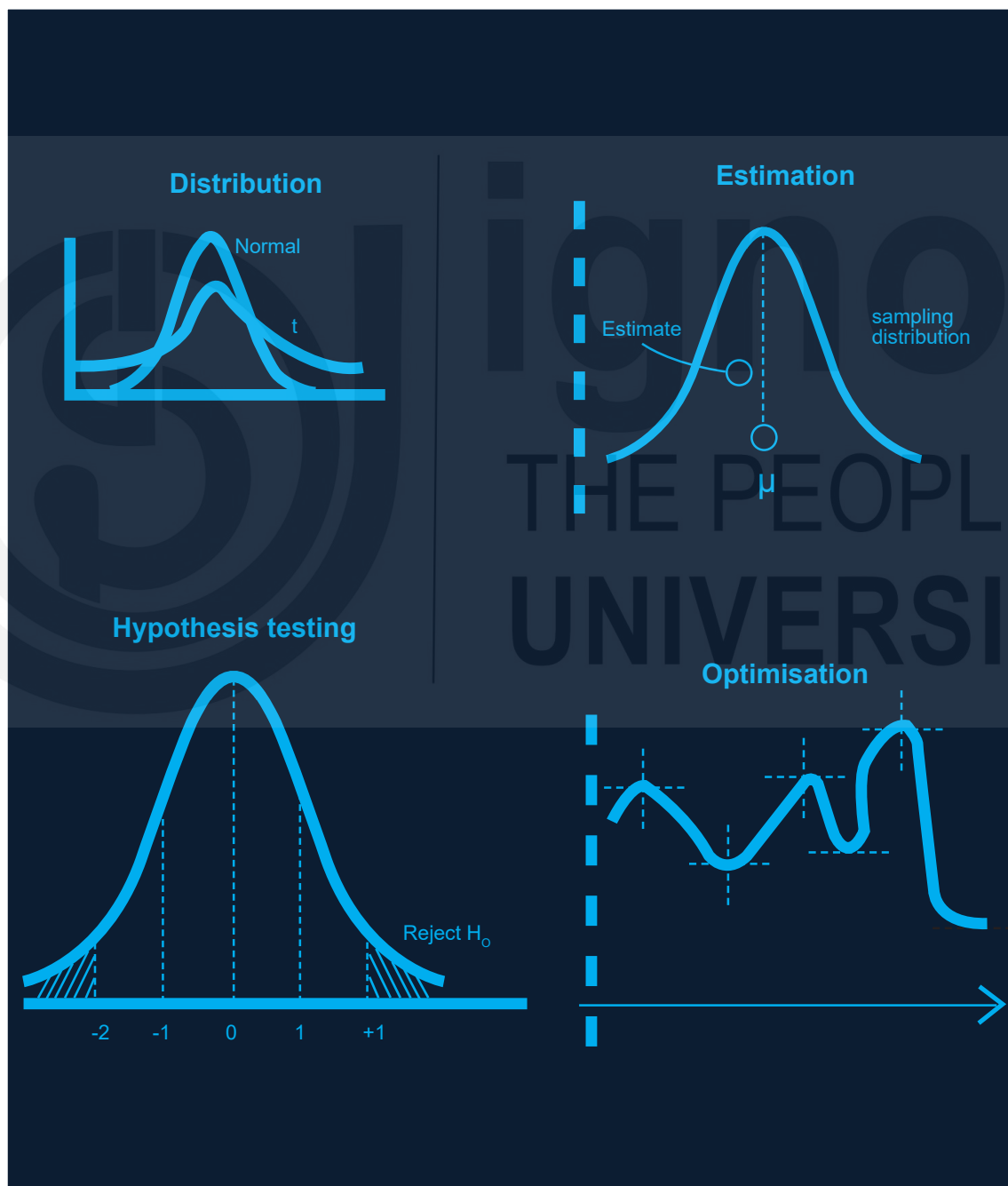


MCS-066

Mathematical Foundations - II



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Unit 2: From MST-001 Unit 13, Unit 15, Unit 16, MST-002 Unit 1

Unit 3: From MST-002 Unit 2, Unit 3, MPC-006 Unit 1

Block 2:

Unit 4: From MST 003, Unit 1, Unit 2, Unit 3, Unit 4

Unit 5: From MST 003, Unit 5, Unit 6, Unit 7, Unit 8

Unit 6: From MST 003, Unit 9, Unit 10, Unit 11

Unit 7: From MST 003, Unit 12, Unit 15, MST-004 Unit 3, Unit 4

Block 3:

Unit 8: From MST 004, Unit 1, Unit 2

Unit 9: From MST 004, Unit 5, Unit 6, Unit 7, Unit 8

Unit 10: From MST 004, Unit 9, Unit 10, Unit 11

Block 4:

Unit 11: From MEV 019, Unit 12

Unit 12: From MST 004, Unit 16, Unit 13

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Appendix: Appendix of MST-016

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COURSE INTRODUCTION

This course introduces the students to the mathematical foundations of data science. This course provides a detailed introduction to important concepts in statistics, Probability, random variables, hypothesis testing, and optimisation, which may be required in the fields of data science and artificial intelligence. The basic objectives of the course are to:

- introduce the basic concepts of descriptive statistics;
- describe the measures used for describing central tendencies of data;
- explain various measures of dispersion of datasets;
- explain the process of computing the probability of statistical events;
- describe the discrete and continuous random variables;
- define different discrete and continuous probability distributions;
- explain the sampling distribution;
- define the joint distribution of the mean;
- demonstrate the use of the process of hypothesis testing;
- describe the use of the ANOVA test;
- demonstrate the use of several tests used for categorical data;
- introduce an iterative technique for optimisation.

The course consists of four blocks.

Block 1 covers the basic concepts of descriptive statistics. It defines the concept of population, sample, measures of central tendency, different kinds of visualisations for summary statistics, etc. This block also discusses measures of dispersion, including variance, standard deviation, skewness, and more.

Block 2 covers details of probability, including the axioms of probability, conditional probability, Bayes' formula, and independent events. This follows the discussion of random variables and the probability distribution of these random variables. This Block also discusses various discrete probability distributions, such as Bernoulli, Binomial, Poisson, and Hypergeometric, as well as continuous probability distributions like uniform and normal, and a few sampling distributions like the Chi-Square, t-distribution, and F-distribution.

Block 3 introduces you to the statistical process of parameter estimation and hypothesis testing. This block first introduces you to various characteristics of the sampling distribution. It also explains how to determine the sample size needed for an estimate. This block also explains the concept of the maximum likelihood estimator. Further, it discusses the point estimation and confidence interval for a normal population. In addition, the unit discusses the concept of hypothesis testing, explaining various types of tests such as the z-test and t-test.

Block 4 focuses on the concepts relating to the analysis of data and optimisation. This Block discusses the analysis of variance (ANOVA) test in detail. In addition, it discusses tests like goodness of fit, test of independence, etc, for categorical data. This block also introduces the concept of optimisation and discusses an important iterative technique for optimising functions.

Further Readings and References: This course material has been adopted from various courses of IGNOU, as indicated in each Unit. Although the material is self-contained and self-explanatory in nature. In case some learners are interested in gaining more and want to study the contents in greater depth/more detail, you may study the references in the IGNOU course material. Further, it is friendly advice for you to put in a lot of practice to attempt all the exercises given in the relevant chapters of the books listed below:

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Block

1

Descriptive Statistics

UNIT 1

Introduction to Statistics

7

UNIT 2

Describing Data Sets and Measures of Central Tendencies

31

UNIT 3

Measures of Dispersion

73

INTRODUCTION TO BLOCK 1

This being the first block of the course, an attempt has been made to explain the descriptive statistics, which describe the properties of the data. The techniques of descriptive statistics are frequently used for analysing data. The purpose of discussing those techniques is to make you aware of the three major properties that describe the data. These properties are:

- The numerical value of an observation (commonly referred to as the central value) around which most of the other numerical values show a tendency to concentrate or group, is called central tendency.
- The degree to which numerical values are scattered or dispersed around the central value is called dispersion.
- The extent of departure of numerical values from symmetrical distribution (taken as normal distribution) around the central value is called skewness.

This block explains these concepts in detail. This block is divided into three units:

Unit 1 defines the terms statistics, measurement scale, the process of data collection and the basic definitions of population, sample, descriptive statistics, inferential statistics and probability models.

Unit 2 focuses on introducing frequency distributions and graphical representations of data. It discusses different types of graphs, including Histograms, Frequency Polygons, Frequency Curves, Cumulative Frequency Curves (Ogives), Boxplots, etc. In addition, the unit discusses different measures of Central Tendency, such as Mean, Median, and Mode. The unit also describes terms such as quartiles, Deciles, and Percentiles.

Unit 3 discusses the measures of dispersion. This unit defines basic measures of dispersion, like Range, Quartile deviation, mean deviation, variance, standard deviation, etc. This unit also introduces the concept of skewness and kurtosis and defines the characteristics of normal data sets.

The content of each unit is adopted from several units, listed in the reference section of each unit.

UNIT 1 INTRODUCTION TO STATISTICS

- 1.0 Introduction
- 1.1 Objectives
- 1.2 Statistics: Definition and Uses
 - 1.2.1 Definition of Statistics
 - 1.2.2 Uses of Statistics
 - 1.2.3 Limitations of Statistics
- 1.3 Measurement Scales
- 1.4 Types of Data
- 1.5 Data Collection
- 1.6 Population and Sampling
- 1.7 Descriptive and Inferential Statistics
 - 1.7.1 Descriptive Statistics
 - 1.7.2 Inferential Statistics
 - 1.7.3 Probability Models
- 1.8 Summary
- 1.9 Solutions/Answers
- 1.10 References

1.0 INTRODUCTION

This is the first Unit of the course on Mathematical Foundations. This Unit introduces you to the basics of statistics, which have been discussed in detail in this course. The unit defines the term statistics and traces a brief history of the development of statistics. The uses of statistics are also highlighted briefly. The statistical analysis is performed on the data, which is obtained by means of data collection processes. This Unit first introduces the data measurement scales and then defines the types of data. It also briefly presents the data collection methods. In addition, the unit also provides a brief introduction to the terms population, population parameters and sample.

After collecting the data, you would like to use this data to perform an analysis. Therefore, a brief introduction has been provided about the terms descriptive analysis and inferential analysis. Since the analysis is based on a probability model, a brief account of probability models has been introduced in this unit.

This is an introductory unit, and some of the concepts discussed in this unit will be explained in the later units.

1.1 OBJECTIVES

After going through this unit, you should be able to:

- define statistics and briefly trace its evolution;
- list and illustrate the uses of statistics;
- explain different measurement scales;
- define different types of data;
- list various methods of data collection;
- differentiate between population and sampling;
- define the terms descriptive and inferential statistics.

1.2 STATISTICS: DEFINITION AND USES

Consider the following case:

An Open University has many regional offices. These regional offices maintain records of fees collected from the students and the expenses made by the office in different programmes using spreadsheet software. These records may be used to determine information such as:

- Fee collection of each programme.
- The percentage of students of a given programme who have not paid the fee.
- Which programme is most profitable?
- Total expenses of each programme.
- Which programmes are not profitable and may be changed?
- Which programme has the maximum number of students?
- Programme-wise distribution of the fee collected.
- Number of dropout students of a programme.
- Expected fee collection of future years.
- Expected number of students in different programmes during the next term.

How can you create such information from the raw data on fee records of the students? You will be using Statistics to do so.

Statistics is used by human society, knowingly or unknowingly, right from the beginning of its existence. This is because (i) most of the decisions taken by a human being are based on past experiences (i.e. based on statistical data he/she has actually experienced), and (ii) future events are also predicted by using/examining the past behaviour of that particular event.

The credit for establishing Statistics as a discipline in India goes to Prasanta Chandra Mahalanobis (P.C. Mahalanobis). He was a professor of physics in the Presidency College in Kolkata. On 17 December 1931, this statistical laboratory was given the name Indian Statistical Institute (ISI) mainly to promote the study, dissemination of knowledge of Statistics, research in it and to develop statistical techniques which play a major role in addressing various problems of planning of national development and social welfare in the country. Some well known personalities other than P.C. Mahalanobis associated to ISI whose research work made the institute unique internationally, are Professor C.R. Rao, Professor R.C. Bose, Professor S.N. Roy, etc.

Statistics in the World

Without going into more details, we will concentrate on only some major discoveries in the area of Statistics at the international level. Some key developments in statistics are:

- John Graunt (1620-1674) has the credit of producing the first life table with probabilities of survival to each age. Due to this great achievement, he is known as the father of vital statistics.
- Gauss (1777-1855) gave the principle of least squares and the normal law of errors.
- J. Bernoulli (1654-1705) was the first person who stated the law of large numbers.
- Statistical methods in the field of biometry were first introduced by Sir Francis Galton (1822-1911).
- Later, Professor Karl Pearson (1857-1936) created correlation coefficient theories. He was also the founder of the Statistical Research Laboratory in the University College, London in 1911. Credit of discovery of Chi-Square test also goes to Karl Pearson.
- Credit of discovery of 't test' or 'student t' test goes to W.S. Gosset, who wrote under the pseudonym of student's 't'.
- Credit of discovering a very powerful test known as Analysis of Variance (ANOVA) goes to Sir Ronald A. Fisher (1890-1962). Fisher also did a lot of work in the area of point estimation. Due to his remarkable contribution in the field of Statistics, he is known as the Father of Statistics.

1.2.1 Definition of Statistics

Statistics is a very old science, and it has developed through the ages. So, it is not surprising that through its long journey, its definitions given by different authors vary. We will present the most recent and broader definitions of Statistics:

“Statistics is a branch of science which deals with the collection, classification, tabulation, analysis and interpretation of data.”

1.2.2 Uses of Statistics

In present times, statistics is regarded as a science having sound techniques of handling huge data and providing valuable conclusions. Today, statistical methods find their application in almost every sphere of human activity, such as economics, commerce, management, information technology, education, planning, banking, insurance sector, medical science, biology, industrial, agriculture, market research, etc. Let us briefly define some of the applications of statistics.

- **Statistics and Industry:** Statistics plays an important role in quality control and production engineering. For example, various control charts are used to maintain a certain quality level and different inspection plans are used in production engineering. You may find the average life of some products, such as an electric bulb, using Statistical methods.
- **Biology and Statistics:** Professor Karl Pearson has stated that the whole doctrine of heredity rests on statistical basis. This is generally said that the height of the child is associated with the height of the father. To test this type of hypothesis, statistics is the only science that provides the scientific methods. Vital statistics is totally devoted to the different aspects of human life, like the average life expectancy of men and women, birth and death rates, etc.

- **Statistics and Medicine:** Statistics also plays an important role in the field of medicine. The hypothesis of the types: (i) Drug A is better than drug B. (ii) Smoking and cancer are associated. (iii) Smoking and TB are associated. All are tested using t-test or χ^2 -test as the case may be. Statistics also find its application in clinical trials.
- **Statistics and Planning:** Every institution/organisation plans for its future targets. Now, a days for good planning, it has become necessary to analyse the statistical data according to the field of interest, such as availability of raw material, consumption, investment, resources available, income, expenditure, quality needed, etc. In order to analyse these types of data, one has to totally depend on the statistical techniques.
- **Statistics and Commerce:** In present times, there is a very tough competition in almost every business. Also, fashions, likings/tastes, requirements, trends, levels of qualities, technologies, etc. are changing very fast. So, for the success of the business, it has become necessary for a businessperson to know the coming trend of the market in advance or as soon as possible. This can also be achieved only with the help of market survey, which requires statistical techniques.
- **Statistics and Agriculture:** Presently, there are a number of varieties of seeds for a particular crop. Also, different types of fertilisers are available in the market. For a good yield, it has become necessary to know which one is better. This job is again done by a very popular and widely used test known as Analysis of variance (ANOVA), discovered by Professor R.A. Fisher.
- **Statistics and Insurance Sector:** Insurance sector totally depends on the statistical data and different concepts of probability theory. Life tables lie at the heart of human insurance. Future lifetime of a life is calculated using the concept of random variables and their expected values. Due to the large use of statistics in insurance sector, a new branch of statistics known as Actuarial statistics has been started in some institutes throughout the world.
- **Statistics and Research:** Research is a very important aspect in every discipline. In many disciplines, such as psychology, tourism, education, management, engineering, etc., one has to collect data on the characteristics of interest under study. Some of the related questions here are: which measurement of scale to be used, and which tests to be used. This requires the knowledge of different types of measurement scales and, accordingly, a suitable statistical tool is needed.
- **Statistics and Economics:** In order to know about the development of a country, it has become necessary to obtain the data related to its economic growth. Again, statistical tools are needed to collect relevant data (such as related to agricultural, industrial, literacy, etc.) and for its analysis.

List of fields/areas where statistics is used does not end here. We have just touched some of the areas where statistics has its application. We close this section by saying that statistics can be used in any field of interest.

1.2.3 Limitations of Statistics

In the previous sub-section of this unit, you have seen a wide range of applications of statistics. Statistics also have its own limitations; some of them are described as follows:

- (1) **Indirect Approach Towards Qualitative Characteristic:** Science of statistics basically deals with numerical data. Therefore, statistical tools are applicable only for quantitative measures. But many times, characteristic under study is qualitative in nature, such as honesty, beauty, intelligence, boldness, drinking, smoking, etc. So, any statistical tool cannot be directly applied on these types of characteristics. However, study of these types of characteristics can be made possible by first converting the characteristic under study into numerical figures based on some uniform criteria. For example, intelligence can be converted into numerical figures with the help of the marks obtained by the individuals in a common test.
- (2) **Dealing with a Group:** The Science of statistics deals with aggregates of objects, not with individuals. The individual's figures, when taken separately, do not come under the category of statistical data. So, applicability of any statistical tool becomes meaningless. For example, salary of one employee of an institute does not give any message related to the salaries of the employees of that particular institution.
- (3) **Lack of Exactness:** Statistical results are not exactly true, but they are true on an average. For example, if a statistical report says that 70% population of India lives in rural areas. It does not imply that if you visit a public place like a bus stand, a railway station, etc. and ask the people about their living place. Results may surprise you and may highly differ from the above figure. But you may note that as sample size increases, the result will also come nearer and nearer to exact figure 70%. But there are sciences like mathematics where exactness is maintained. For example, if a book seller gets 5% profit on selling a particular book. Then it is sure that if the sale price of that particular book is Rs 200, he/she will get Rs 10 as profit, and in case of the sale price of Rs 300, the profit will be Rs 15, and so on.
- (4) **Requirement of Experts' Hands for Effective Use:** Requirement of experts' hands for effective and appropriate use is one of the main drawbacks of the science of statistics. There are many statistical tools of similar type. For example, to find average in a particular situation, which of the possible tools like mean, median, mode, geometric mean, harmonic mean, etc., is appropriate, needs the hands of experts. Similarly, to test a given statistical hypothesis, which of the possible tools like Z-test, t-test, χ^2 -test, F-test, ANOVA, median test, run test, sign test, etc. is appropriate, again needs the hands of experts. This limitation of the statistics limits the range of its effective users.

After going through the basics of statistics, let us now discuss an important aspect of statistics – measurement in the next section.

1.3 MEASUREMENT SCALES

Two words, “counting” and “measurement”, are very frequently used by everybody. For example, if you want to know the number of pages in a notebook, you can easily count them. Also, if you want to know the height of a man, you can easily measure it. But, in Statistics, act of counting and measurement is divided into four levels of measurement scales known as

1. Nominal Scale
2. Ordinal Scale
3. Interval Scale
4. Ratio Scale

Let us discuss these scales of measurement one by one:

Nominal Scale

The word nominal has come from this Latin word, i.e. ‘Nomen’. In Latin, ‘Nomen’ means name. Therefore, under nominal scale, we divide the objects under study into two or more categories by giving them unique names. The classification of objects into at least two or more categories is done in such a way that

- (a) Each object takes place only in one category, i.e. each object falls in a unique category, i.e. it either belongs to a category or not. Mathematically, we may use the symbol (“=”, “ $\neq$ ”) if an object falls in a category or not.
- (b) Number of categories must be sufficient to include all objects, i.e. there should not be scope for missing even a single object, which does not fall in any of the categories. That is, in statistical language, categories must be mutually exclusive and exhaustive.

Generally nominal scale is used when we want to categorise the data based on a characteristic, such as gender, race, region, religion, etc.

To get more familiar with the idea of nominal scale, let us consider some examples:

(i) Classification into Different Categories Based on Gender

This can be done by dividing the population into three categories: male ‘M’, female ‘F’ and Third Gender ‘T’

Category	Name/Code
Male	M or 0
Female	F or 1
Third Gender	T or 2

Please note that the main thing is that we have to give a unique name to each category.

(ii) Classification into Different Categories Based on Region

28 states and 7 union territories together classified India into 35 categories, which can be coded by their usual names or may be coded by using some other symbols.

Note 1: We note that in nominal scale, we have just coded the objects. Signs of less than or greater than do not make any sense in a nominal scale. For example, Female > Male or Male > Female does not make any sense. That is, we cannot talk about the order between the three categories in the case of a nominal scale.

If in a measurement scale, order also makes sense, then this scale comes under the heading of an ordinal scale, discussed below.

Ordinal Scale

We have seen that order does not make any sense in nominal scale. As the name ordinal itself suggests that other than the names or codes given to the different categories, it also provides the order among the categories. That is, we can place the objects in a series based on the orders or ranks given by using ordinal scale. But here we cannot find actual difference between the two categories.

Generally, ordinal scale is used when we want to measure the attitude scores towards the level of liking, satisfaction, preference, etc. Different designations in an institute can also be measured by using an ordinal scale.

To get more familiar with the concept of ordinal scale, let us consider some examples:

- (i) Opinion of persons about proposal of introducing co-education in a college comes under this scale. Suppose we assign '1' to strongly disagree, '2' to disagree, '3' to indifferent (or neutral), '4' to agree and '5' to strongly agree. Here, the order also matters, and mathematically we may use the symbols $>$, $<$ in addition to those used for nominal scale, i.e. $=$, $\neq$ as here strongly agree opinion comes first in order as compared to agree and so on, i.e. $5 > 4 > 3 > 2 > 1$ or $1 < 2 < 3 < 4 < 5$. But actual difference between different categories in this Likert Scale is not possible. This is because, suppose "strongly agree" means he/she gives marks from 75% to 100% for the co-education to be introduced and suppose "agree" means the marks are given in the range, say 50% to 75%. Now, the actual difference between "strongly agree" and "agree" is not feasible in this sense.
- (ii) Based on the income of a family, generally, families of a society may be divided into three categories:

Higher income group

Middle-income group

Low-income group

The income of a person in the higher income group is better than that of a person in the middle income group, and the income of a person in the

middle income group is higher than that of a person in the lower income group. But the actual difference between them cannot be measured.

In all the above examples, the actual difference is not possible because all the ranks are on the ranges and not on fixed points.

Now, we are going to study the next higher level of measurement, wherein the actual differences can be found. This scale is known as interval scale.

Interval Scale

The third level of measurement, i.e. interval scale, possesses the property of difference, which was not satisfied in the case of nominal and ordinal scales. The interval scale is used when we want to measure years/historical time/calendar time, temperature (except in the Kelvin scale), sea level, marks in the tests where there is negative marking, etc. Mathematically, this scale includes $+$, $-$ in addition to $>$, $<$ and $=$, $\neq$, $\neq$.

To get more familiar with the concept of interval scale, let us consider some examples:

- (i) Measurement of temperature in degrees Celsius ($^{\circ}\text{C}$ assumes 0°C when water starts freezing to ice. So, in degrees Celsius, the origin is arbitrary; that's why measurement of temperature in degrees Celsius comes in an interval scale. Because the degree Celsius origin is arbitrary, we cannot say that 30°C is twice as hot 15°C . Similarly, measurement of temperature in Fahrenheit comes in the interval scale.
- (ii) Mean sea level (MSL) also has an arbitrary origin because it is the mean of two means, mean high tide and mean low tide (and mean high tide and mean low tide vary according to high and low pressure zones). Further, it also varies from place to place and time to time. So, measurement of sea level also comes in the interval scale.

Ratio Scale

Ratio scale is the highest level of measurement because nominal scale gives only names to the different categories, ordinal scale provides orders between categories other than names, interval scale provides the facility of difference between categories other than names and orders but ratio scale other than names, orders and characteristic of difference also provides natural zero (absolute zero). In ratio measurement, scale values of a characteristic cannot be negative.

Ratio scale is used when we want to measure temperature in Kelvin, weight, height, length, age, mass, time, plane angle, etc. Ratio scale includes $\times$, $\div$, $\times$, $\div$ in addition to $+$, $-$, $>$, $<$, $=$, $\neq$. But be careful never to take '0' in the denominator while finding ratios. For example, $4/0$ is meaningless.

To get more familiar with the concept of ratio scale, let us consider some examples, where ratio scale is used:

- (i) Measurement of temperature in Kelvin scale comes under ratio scale because it has an absolute zero, which is equivalent to -273.15°C . This characteristic of origin allows us to make the statement that 50K ('50K' read as 50 degrees Kelvin) is 5 times hotter compared to 10K.

- (ii) Measurement of money also comes under the ratio scale because it satisfies all the requirements of an interval scale and has a natural zero. For example, suppose person A has Rs 100 and person B has Rs 500, then we can say that B has 5 times the amount of A. Also, a person may not have any money (i.e. absolute zero '0'). Here, the natural zero indicates the absence of money for a person. Thus, it satisfies all the requirements of a ratio scale.
- (iii) Both height (in cm.) and age (in days) of students of a particular university satisfy all the requirements of a ratio scale. Because height and age both cannot be negative (i.e., have an absolute zero).

Permissible Statistical Tools

One of the advantages of measurement scale is that these help us to decide which statistical tool should be used in a given situation. Table 1.1 shows the list of permissible statistical tools in the case of nominal, ordinal, interval and ratio scales. Based on information provided by these scales, their levels from lowest to highest are nominal, ordinal, interval and ratio. That is why all the Statistical tools applicable on the lower scale will automatically be applicable on the next level scale. So, we will not repeat the permissible statistical tools used in lower level scale.

Table 1.1: The Measurement Scale and Permissible Statistical Tools

MEASUREMENT SCALE	PERMISSIBLE STATISTICAL TOOLS	LOGIC/REASON
NOMINAL SCALE	Mode, chi-square test	Here counting is only permissible operation.
ORDINAL SCALE	Median all positional averages like quartile, Decile, percentile, Spearman's Rank correlation.	Here other than counting, order relation (less than or greater than) also exists.
INTERVAL SCALE	Mean, S.D., t-test, F-test, ANOVA, sample multiple and moment correlations, regression.	Here counting, order and difference operations hold.
RATIO SCALE	Geometric mean (G.M.), Harmonic mean (H.M.), Coefficient of variation.	Here counting, order, difference and natural zero exist.

Check Your Progress 1

1. Is machine learning or deep learning an application of statistics?

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2. At a picnic spot in India, 1000 tourists visit over a period of 7 days. Each tourist is asked the name of the country in which he/she was born.

Then, the data thus obtained comes under which measurement scale?
Give reasons in support of your answer.

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3. Answer the following questions:

- (i) Which scale is at the lowest level?
- (ii) Which scale is at the highest level?
- (iii) Which scale has absolute zero?
- (iv) Which scale is used to find the mean sea level (MSL)?

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4. Answer the following questions:

- (i) In which scale is the median not permissible?
- (ii) In which scale(s) mean is not permissible?
- (iii) In which scale(s) are geometric mean and harmonic mean not permissible?
- (iv) In which scale are geometric mean and harmonic mean permissible?

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1.4 TYPES OF DATA

Data plays the role of raw material for any statistical investigation. Data can be defined in a single sentence as:

“The values of different objects collected in a survey or recorded values of an experiment over a time period taken together constitute what we call data in Statistics”

Each value in the data is known as an observation. Statistical data based on the characteristic, nature of the characteristic, level of measurement, time, and ways of obtaining it may be classified as follows:

- Quantitative data
 - Qualitative data
- } based on the characteristic
-
- Discrete data
 - Continuous data
- } based on nature of the characteristic
-
- Nominal data
 - Ordinal data
 - Interval data
 - Ratio data
- } based on level of measurement
-
- TimeSeries data
 - Cross Sectional data
- } based on time component
-
- Primary data
 - Secondary data
- } based on the ways of obtaining the data
-
- Quantitative data
 - Qualitative data
- } based on the characteristic
-
- Discrete data
 - Continuous data
- } based on nature of the characteristic
-
- Nominal data
 - Ordinal data
 - Interval data
 - Ratio data
- } based on level of measurement
-
- TimeSeries data
 - Cross Sectional data
- } based on time component
-
- Primary data
 - Secondary data
- } based on the ways of obtaining the data

Let us discuss different types of data one by one:

Quantitative Data

As the name quantitative itself suggests that it is related to quantity. In fact, data are said to be quantitative data if a numerical quantity (which exactly measures the characteristic under study) is associated with each observation.

Generally, interval or ratio scales are used as a measurement scale in case of quantitative data. Data based on the following characteristics generally gives quantitative type of data, such as weight, height, age, length, area, volume, money, temperature, humidity, size, etc.

For example,

- (i) Weights in kilograms (say) of students in a class.
- (ii) Height in centimetres (say) of the candidates appearing in a physical examination.
- (iii) Length (in cm) of different tables in a furniture showroom.

Qualitative Data

As the name qualitative itself suggests that it is related to the quality of an object/thing. It is obvious that quality cannot be measured numerically in exact terms. Thus, if the characteristic/attribute under study is such that it is measured only on the basis of presence or absence, then the data thus obtained is known as qualitative data.

Generally, nominal and ordinal scales are used as measurement scales in case of qualitative data. Data based on the following characteristics generally gives qualitative data, such as gender, marital status, qualification, colour, religion, satisfaction, types of trees, beauty, honesty, etc.

For example,

- (i) If the characteristic under study is marital status, then objects may be divided into four categories: married, unmarried, divorced, and widowed.
- (ii) If the characteristic under study is qualification (say) 'matriculation', then objects can be divided into two categories as 'Matriculation passed' and 'Matriculation not passed'.

Discrete Data

If the nature of the characteristic under study is such that values of observations may be at most countable between two limits, then corresponding data are known as discrete data.

For example,

- (i) The number of books on the shelf of an almirah in a library forms discrete data, as the number of books may be 0, 1, 2, 3 and so on. This is because the number of books cannot take fractional values.
- (ii) If there are 30 students in a class, then number of students present in a lecture forms discrete data.

Continuous Data

Data are said to be continuous if the measurement of the observations of a characteristic under study may be any real value between two limits.

For example,

- (i) Data obtained by measuring the heights of the students of a class of 30 students form continuous data, because if the minimum and maximum heights are 152cm and 175 cm, respectively, then the height of a student may take any possible value between 152 cm and 175 cm. For example, it may be 152.2 cm, 160.3 cm, etc.
- (ii) Data obtained by measuring weights of the students of a class also form continuous data because weights of students may be 48.235kg, 50.270kg, 42.330kg, etc.

You can refer to the scales of measurement in Section 1.3 for details on nominal, ordinal, ratio and interval types of data

Time Series Data

The collection of data is done to address a specific purpose. The purpose may have its connection with time, geographical location or both. If the purpose of data collection has a connection with time, then it is known as time series data. That is, in time series data, time is one of the main variables and the data is collected usually at regular intervals of time related to the characteristic(s) under study. The objective here is to show how characteristic(s) change over time.

For example, quarterly profit of a company for last eight quarters, yearly production of a crop in India for last six years, yearly expenditure of a family on different items for last five years, weekly rate of inflation for last ten weeks, etc. all form time series data.

Yearly expenditures (in Rs) for a family on different items from 2006 to 2010, as given in the following table, are nothing but time series data.

Year	Food	Education	Rent	Miscellaneous	Total
2006	40000	10000	36000	20000	106000
2007	45000	12000	40000	28000	125000
2008	54000	15000	45000	32000	146000
2009	60000	20000	50000	40000	170000
2010	70000	30000	55000	45000	200000

Note 1: If the purpose of the data collection has its connection with geographical location, then it is known as **Spatial Data**.

For example,

- (i) Price of petrol in Delhi, Haryana, Punjab, Chandigarh at a particular time.
- (ii) Number of runs scored by a batsman in different matches in a one day series in different stadiums.

Note 2: If the purpose of the data collection has its connection with both time and geographical location, then it is known as **Spatio-Temporal Data**.

For example, data related to population of different states in India in 2001 and 2011 will be Spatio-Temporal Data.

Cross Sectional Data

Sometimes we are interested to know that how a characteristic (such as income or expenditure, population, votes in an election, etc.) under study at one point in time is distributed over different subjects (such as families, countries, political parties, etc.). This type of data, which is collected at one point in time, is known as cross sectional data.

For example, annual income of different families of a locality, survey of consumer's expenditure conducted by a research scholar, opinion polls conducted by an agency, salaries of all employees of an institute, etc.

Note:

- (i) If you are interested to know the changes in a characteristic, say expenditure of a family over a period of time, then you have to use time series data.
- (ii) If you are interested to know the changes in a characteristic, say expenditure of different families at single point in time, you have to use cross sectional data.

Primary Data

Data which are collected by an investigator or agency or institution for a specific purpose and these people are the first to use these data, are called primary data. That is, these data are originally collected by these people and they are the first to use these data.

For example, suppose a research scholar wants to know the mean age of students in class 10 of a school. S/he collects data related to the age of each student in class 10 of that school by contacting each student personally, then the data obtained by the research scholar is an example of primary data for the same research scholar.

Secondary Data

Data obtained/gathered by an investigator or agency or institution from a source which already exists, are called secondary data. That is, these data were originally collected by an investigator or agency or institution and have been used by them at least once. And now, these data are going to be used at least second time

For example, consider the same example as discussed in case of primary data. If the research scholar collects the age data of the students from the records of that particular university, then the data thus obtained is an example of secondary data. Note that, in both the cases data remains the same, only way of collecting the data differs.

1.5 DATA COLLECTION

The data collection is performed to create primary data. There are a number of methods of collection of primary data depending upon many factors such as geographical area of the field, money available, time period, accuracy needed, literacy of the respondents/informants, etc. Here we will discuss briefly only the following commonly used methods.

- (1) Direct Personal Investigation Method
- (2) Telephone Method
- (3) Indirect Oral Interviews Method
- (4) Local Correspondents Method
- (5) Mailed Questionnaire Method

Let us discuss these methods briefly.

(1) Direct Personal Investigation Method

In this method, the investigator personally contacts the informants and collects the related data through face to face interviews of the informants. Due to face to face meeting of investigator and informants data collected under this method has maximum degree of accuracy. But the degree of accuracy depends upon the sincerity, honesty, unbiasedness and expertness of the investigator, because it is the investigator who ultimately gives the final shape to the information provided by the informants. This method of collection of primary data is recommended/suitable when field of enquiry is small, secrecy related to data is need, high accuracy is required and time as well as money is sufficient.

(2) Telephone Method

In the direct personal investigation method investigator has to personally contact the informants, but now-a-day's telephone is a very good communication tool. If the information of the interest is collected through telephone then data so obtained come under telephone method.

(3) Indirect Oral Interview Method

In this method, investigator does not meet the actual informants directly, but the related information is collected/obtained from other persons who are supposed to have the required type of information. The informants who provide the information about actual informants are known as 'witnesses'.

The success of this method mainly depends upon the skill and experience of the investigator. Because it depends upon the way, sequence and trick of questions prepared and asked by the investigator from the informants. It also depends on the behaviour and how much he/she is capable to create confidence in the informants.

For example, if a researcher wants to collect the data on the smoking habit of students of a particular class of a college, then it may happen that actual informants do not provide correct information. So, data can be collected with the help of class met or college met or from the neighbours.

(4) Local Correspondents Method

In this method, first of all some local correspondents or agents are appointed by the investigator or agency or institute to collect data. These local agents directly meet to the informants and collect data related to the required purpose in hand. Data collected by these local agents has a high degree of accuracy, because they are familiar with the local language, traditions, general behaviour of the people, etc., of that particular area.

This method of collection of primary data is suitable if

- Area of investigation is large, e.g. news channels have their reporters throughout India.
- Time period in which information is needed is very short. For example, news related to the happening of any special incident can be easily seen on the news channels in very short time period after its happening.
- Information is needed on a regular basis. For example, news are provided daily using this method.

(5) Mailed Questionnaire Method

In this method, first of all, a list of questions related to the information required by the investigator is prepared. At the time of preparing the list of questions, the following points should be kept in mind.

- Number of questions should not be too many.
- Each question should be related directly or indirectly to the objective(s) of the investigation.
- Each question should be clear.
- Generally, objective type questions should be used, but if necessary, multiple choice and open-ended questions can also be used.
- Language should be simple and effective.

After preparing the 'final list' of questions known as questionnaire, it is sent through mail to the informants. A covering letter requesting informants to fill the questionnaire and return it, and a brief introduction about the objective of the investigation, is also attached. This method is suitable when the informants are literate and area of investigation is large.

Check Your Progress 2

1. Characterise the following data as qualitative or quantitative data?

- (i) Cities of a country
- (ii) Temperature
- (iii) Attitude towards learning
- (iv) Fee details

.....

2. Identify whether the data are discrete or continuous in the following cases:

- (i) Number of people present at a party.
- (ii) Length of leaves of a plant.
- (iii) Lifetime in hours of an electrical bulb.
- (iv) Number of cars standing in a showroom over a period of 7 days.
- (v) Number of patients who visited a hospital on a particular day.

.....

3. Give reasons in each case whether the data are primary or secondary.

- (i) 2011 census data published by the Office of the Registrar General of India and to be used by itself.
- (ii) 2011 census data to be used by a demographer in his study.

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4. You have to collect data from about 5000 students of various colleges relating to their experiences in Learning. Which data collection method would you use? Justify your answer.

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1.6 POPULATION AND SAMPLING

One of the important considerations while collecting primary data is identifying the population and collecting a representative sample from this population. In this section, we briefly discuss population and sampling.

Population: Population literally means a well-defined group (collection or bunch) of some objects (units or elements). Examples of populations and its units are: a city with some clear-cut territory having its dwellers as units (elements) or having houses as its units; a hospital with indoor patients as units or its doctors as units; a library with its employees as units or books as its units; a school with enrolled students as units; and so on. Thus, population is the collection or group of individuals /items/ units/ observations under study. The total number of elements/items/units/observations in a population is known as its size and generally denoted by N .

Sample: A finite subset of units of a population is called a “sample” and the number of units belonging to the sample is called the sample size. If n denotes the sample size, then necessarily we have the condition $n < N$ and then the sample is said to be a proper sample. However, in some situations, we have $n = N$, that is, all the units of the population are selected as the sample. In almost all kinds of statistical studies, a sample is selected for inferential purposes. However, even if the population is not large enough or it has a well-defined territory or it is not of destructive type; a sample is useful since besides it is always less costly, less time consuming, it helps us to get a handy result in very short time.

A sample of size n is not selected simultaneously, rather n units are selected one by one in n draws in order to constitute a sample. The process through which units of the population are selected in different draws to constitute a sample of specified size, is known as “sampling”. Thus, sampling is a draw-to-draw mechanism of selecting some units from a population. Sampling is mainly of two types, namely, (a) Probability Sampling or Random Sampling, and (b) Non-Probability Sampling. The technique of random sampling is of fundamental importance in the application of statistics. The Estimation theory is based on the assumption of random sampling. It is a scientific method of selecting samples accordingly to some laws of chance in which each unit in the population has some definite pre-assigned probability of being selected in the sample. Simple Random Sampling, Stratified Sampling, Systematic Random Sampling, etc., are examples of random sampling. In non-probability sampling, the sample is selected with definite purpose and, hence, the choice of sampling units depends entirely on the discretion and judgment of the investigator. It is, therefore, not a scientific method and attracts severe criticisms in many circumstances. Purposive Sampling, Judgement Sampling, etc., are the

examples of non-probability sampling. We know that if $n < N$, it is a sampling procedure. When $n = N$, the process is known as “complete enumeration or census”. An example of complete enumeration is the decennial census of India.

Need for Sampling in Statistical Inference: In many practical and real situations, the population under consideration is either infinitely large in size or it is unbounded in the sense that its boundaries are not well-defined, or it is of a destructive nature. For example, in order to determine average life of two hundred produced electric bulbs, it would be necessary to light these bulbs unless they all get fused, resulting in destruction of the entire lot of production. Similarly, in order to estimate the proportion of smokers in a large city, information has to be gathered from each and every person dwelling in the city which would be a very difficult task in terms of manpower, money and time to be required. Sometimes, the population size is not known if its boundaries are not well-defined, like number of fish in a pond or lake. In all such cases, it is not feasible to gather information on all the units, rather it becomes almost impossible. Keeping in view the difficulty in contacting each and every unit of the population due to these reasons and also to save the time, money and manpower to be required for this, generally a part of the population, popularly known as a “sample” is selected in some pre-assigned manner which in turn is used for drawing the inferences on the population itself or on the parameters of the population. The results obtained from the sample are projected in such a way that they are valid for concluding about the entire population. Therefore, the sample works like a “Vehicle” to reach (drawing) at valid conclusions about the population. In fact, a sample helps us to reach to the “whole”(population) from a “part” (sample) in all types of statistical studies. Thus, statistical inference can be defined as: “It is a process of concluding (projecting or inferring) something desired about a given population on the basis of sample results”.

Parameter and Statistic: A parameter is a function of population values that is used to represent certain characteristics of the population. For example, population size, population mean, population variance, population coefficient of variation, population proportion, population correlation coefficient, etc., are all parameters since their calculation involve all the population values. A statistic is a function of sample values only and does not contain any unknown population parameter in it. For example, if $X_1, X_2, \dots, X_n$ represent values of the variable X in a random sample of size n taken from a population, then sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is a statistic.

1.7 DESCRIPTIVE AND INFERENTIAL STATISTICS

People use statistics to summarise data and to make decisions about the problems on the basis of different types of information available. The science of statistics may be broadly studied under two headings: i) Descriptive Statistics, and (ii) Inferential Statistics

1.7.1 Descriptive Statistics

After collecting the information in terms of data, we may like to arrange the collected data in a proper manner because the collected data may be huge in volume. So, here the need for proper arrangement of data arises. In statistical terminology, the proper arrangement of data is known as the presentation of data. There are some frequently used tools available for representing data, and they may be classified into three basic forms: statistical tables or tabulations of data, diagrams and graphs. A pictorial presentation of the tabulated data may be done either with the help of different kinds of diagrams or by graphs. Some of these graphs include the graphs for frequency distributions, graphs for time series data, stem-and-leaf displays and box plots.

The next step is to proceed with data analysis to explore the properties of the data. The statistical tools, which describe the properties of the data, are known as descriptive statistics. The techniques of descriptive statistics are frequently used for analysing the data in various fields. The descriptive techniques involve three major properties that describe the data. These properties are:

- The numerical value of an observation (commonly referred as central value) around which most of the other numerical values show a tendency to concentrate or group, called central tendency.
- The degree to which numerical values are scattered or dispersed around the central value, called dispersion.
- The extent of departure of numerical values from symmetrical distribution (taken as normal distribution) around the central value, called skewness.

You will learn more about these in the subsequent units of this block.

1.7.2 Inferential Statistics

The description or distribution of observations is not the prime objective of any scientific investigation. Organising and summarising data is only one step in the process of analysing the data. In any scientific investigation, either the entire population or a sample is considered for the study. In most of the scientific investigations, a sample, a small portion of the population under investigation, is used for the study. On the basis of the information contained in the sample, we try to draw conclusions about the population. This process is known as statistical inference. Statistical inference is widely applicable in behavioural sciences. Inferential statistics deals with drawing conclusions about a large group of individuals (population) on the basis of observation of a few participants from among them or about the events which are yet to occur on the basis of past events. It provides tools to compute the probabilities of future behaviour of the subjects. Inferential statistics is the mathematics and logic of how this generalisation from sample to population can be made.

There are two types of inferential procedures:

Estimation: In estimation, inference is made about the population characteristics on the basis of what is discovered about the sample. There may be sampling variations because of chance fluctuations, variations in sampling techniques, and other sampling errors. A good statistical estimator should be unbiased, consistent and accurate.

Hypothesis Testing: In hypothesis testing, we hypothesise a value that characterises the population of observations. The question is whether that hypothesis is reasonable in the light of the evidence from the sample. Sometimes, hypothesis testing is referred to as a statistical decision-making process.

You will learn more about these in the Block 3 of this course.

1.7.3 Probability Models

As defined in the previous section, you use inferential statistics to make logical conclusions, as determined by data. However, these conclusions are probabilistic in nature. The probability models form the backbone of making such decisions. These probabilistic decisions are either in a certain approximate range or accuracy, say 95%. Thus, an underlying probability model must be considered while making the decisions.

What is a probability model? A probability model mathematically quantifies the likelihood of a prediction or a decision, given the uncertainty of data or processes that are used to make the prediction.

A probability model is defined by three elements: the sample space of a random process, an event of interest, which should be a subset of the sample space and a set of rules of a function that computes the probability for the event of interest. For example, consider that you perform a random experiment of tossing a coin and the outcome of this toss is to be predicted, then the following would define its probability model:

Sample Space is {Heads, Tails}, as a toss of a coin will result in one of the two states.

Assume that you are interested in the event that “Toss of the coin results in Heads.”

The probability function in this case would define the probability as $1/2$, if the coin is fair.

The probability models can be discrete if the integral outcomes are countable and finite, or continuous if the outcomes are represented using real numbers and are uncountable. You will learn more about probability in the Block 2 of this course.

Check Your Progress 3

1. What should you use, population or sample, for the following situations?

(i) Find the number of patients visiting a hospital in a day.

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(ii) Find the attitude of patients towards health.

.....

(iii) Average height of students of class 10.

.....

(iv) Average marks of the students of a School.

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2. A survey was performed to find the average age of a person who performs morning exercises. A sample of 2000 was taken for this. The survey showed that the average age of these people was 48 years. In addition, the researcher stated that her data shows that the national average is expected to be 45 to 51 years. Which of these is a Statistic, and which of the data is a parameter?

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3. Which of the data in question 2 is obtained using descriptive statistics and which one is obtained through inferential statistics?

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1.8 SUMMARY

This Unit defines the term statistics and discusses applications of statistics in different areas. It also highlighted the shortcomings of statistics. Next, the unit explains different measurement scales in detail with the help of examples. Further, the unit explains different kinds of data, such as qualitative, quantitative, time series data, cross-sectional data, and primary and secondary data. The data types will be useful to you, when you study the later units. Next, the unit discusses different ways of collecting primary data. Some of the popular methods are discussed very briefly in this section. As you know about the method of data collection, the unit also discusses about the terms population, population parameters and sampling.

Once the data has been collected, you need to analyse that data. Thus, the unit briefly defines the terms descriptive data analysis, which defines the process of data organisation and summarisation. This is followed by a brief discussion on inferential statistics that may be used to predict the future data or requirements. Finally, unit relates the process of analysis with probability models and explains in brief the concept of a probability model.

You should go through the references for more details on these topics. These references can be downloaded from eGyankosh.ac.in website.

1.9 SOLUTIONS/ANSWERS

Check Your Progress 1

1. Yes, Machine learning uses and builds statistical models, while topics like neural networks in Deep learning use statistical principles.
2. Nominal scale, because the characteristic 'name of the country' divides the tourists into different categories, each labelled with the name of his/her country.
3. (i) Nominal scale is at the lowest level, because it has only one permissible operation, counting.
 (ii) Ratio scale is at the highest level, because it has all four operations: counting, order, distance and absolute zero.
 (iii) Ratio scale is only one of the four measurement scales, nominal, ordinal, interval and ratio scales, which has absolute zero.
 (iv) Because sea level has no absolute zero, so interval scale is used to find the mean sea level (MSL).
4. (i) In order to find median, we have to arrange the data in ascending or descending order of magnitude. But in nominal scale order operation is not present. So, in case of nominal scale data, median is not permissible.
 (ii) In order to find mean, each observation of the data must be associated with a numerical quantity (which exactly measure the quantity of the characteristic). But this requirement is not fulfilled by nominal and ordinal scales data. So, mean is not permissible in case of nominal and ordinal data.
 (iii) In order to find geometric mean (G.M.) and harmonic mean (H.M.) absolute zero must be defined so that one can talk of quotient/ratio of two numbers. But as absolute zero is defined only in ratio scale, so G.M. and H.M. are not permissible in nominal, ordinal and interval scales data.
 (iv) As discussed in (iii), G.M. and H.M. are defined only in ratio scale.

Check Your Progress 2

1

- | | |
|---------------------------------|--------------|
| (i) Cities of a country | Qualitative |
| (ii) Temperature | Quantitative |
| (iii) Attitude towards learning | Qualitative |
| (iv) Fee details | Quantitative |

2.

- (i) Number of people present at a party:

Discrete

(ii) Length of leaves of a plant:

Continuous

(iii) Lifetime in hours of an electrical bulb:

Continuous

(iv) Number of cars standing in a showroom over a period of 7 days:

Discrete

(v) Number of patients who visited a hospital on a particular day:

Discrete

3.

(i) Primary data, as data is used by the office itself

(ii) Secondary data, as data is used by another person, who is not the data creator

4. Mailed questionnaire method, as this method is suitable when the informants are literate and the area of investigation is large. Also, make a questionnaire that is not too large.

Check Your Progress 3

1.

(i) Find the number of patients visiting a hospital in a day: Population, as just one hospital, and limited quantitative data

(ii) Find the attitude of patients towards health: Sample, as very large data and qualitative question

(iii) Average height of students of class 10: Sample, as large data, through quantitative but expensive to use population

(iv) Average marks of the students of a School: Population, limited quantitative data.

2. The average age of these people was 48 years is a Statistic, and the national average is expected to be 45 to 51 years is a parameter.

3. The average age of these people was 48 years (a Statistic), obtained through descriptive analysis, whereas the national average is expected to be 45 to 51 years (a parameter), obtained through inferential statistics.

1.10 REFERENCES

IGNOU Course Material

1. Post-Graduate Diploma In Applied Statistics (PGDAST), MST 001: Matrices, Determinants and Collection of Data, UNIT 11: Introduction to Statistics (for Sections 1.2 to Section 1.4 and related and related portions in Sections 1.0, 1.1, 1.8, 1.9 and Check Your Progress questions)

Descriptive Statistics

2. Post-Graduate Diploma In Applied Statistics (PGDAST), MST 001: Matrices, Determinants and Collection of Data, UNIT 12: Collection and Scrutiny of Data (for Section 1.5 and related and related portions in Sections 1.0, 1.1, 1.8, 1.9 and Check Your Progress questions)
3. MPC-006 Block 1 Introduction to Statistics UNIT 2 Descriptive and Inferential Statistics (for Sections 1.7 and related and related portions in Sections 1.0, 1.1, 1.8, 1.9 and Check Your Progress questions)
4. MEV-019 Research Methodology for Environmental Science, Block 3: Statistical Analysis, Unit 9: Sampling Distributions (for Sections 1.6 and related and related portions in Sections 1.0, 1.1, 1.8, 1.9 and Check Your Progress questions)

Reference Book

5. “Introduction to Probability and Statistics for Engineers and Scientists”, 3rd Edition, Sheldon M. Ross, Elsevier Academic Press, 2004



UNIT 2 DESCRIBING DATA SETS AND MEASURES OF CENTRAL TENDENCY

Structure

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2.0 INTRODUCTION

In Unit 1 of this block, we have defined the concept of statistics, its uses, types of data, data creation, etc. In this unit, we present how you can tabulate and represent data graphically. In this unit, we are going to discuss about the basic analysis of data. We first define the frequency distribution, which aims at classifying measurable data. This data can then be represented using graphs that communicate the characteristics of the data more effectively. Further, for the purpose of analysis, a very important and powerful tool is a single average value that represents the entire mass of data.

The term average in Statistics refers to a one figure summary of a distribution. It gives a value around which the distribution is concentrated. For this reason, the average is also called the measure of central tendency. The calculation of the average condenses a distribution into a single value that is supposed to represent the distribution. This helps both individual assessments of a distribution as well as in comparison with another distribution.

This unit comprises of frequency distribution and its graphical representations. It various measures of central tendency and concepts and methods of calculating the partition values.

2.1 OBJECTIVES

After studying this unit, you would be able to

- define different types of frequency distribution;
 - make graphs like histogram, ogive, stem and leaf plot;
 - explain the measure of central tendency;
 - calculate the different types of measures of central tendency;
 - describe the methods of calculation of partition values;
 - make a boxplot using the partition values.
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2.2 FREQUENCY DISTRIBUTION

A frequency distribution refers to the data which are classified on the basis of some variables that can be measured such as wages, age of children, etc. A variable refers to the characteristic that varies in magnitude in a frequency distribution. It may be either discrete or continuous. A discrete variable is one that generally takes integer values. For example, the number of students, the number of books, etc. A continuous variable can take integer or fractional values within the range of possibilities, such as the height or weight of individuals. Generally speaking, continuous data are obtained through measurements while discrete data are derived by counting. A series described by a continuous variable is called continuous series. Similarly, series represented by a discrete variable is called discrete series.

According to the nature of the variable, the frequency distribution may be of two types, i.e. discrete frequency distribution and continuous frequency distribution.

Discrete Frequency Distribution: A frequency distribution in which the information is distributed in different classes on the basis of a discrete variable is known as a discrete frequency distribution.

Continuous Frequency Distribution: A distribution in which the information is distributed in different classes on the basis of a continuous variable is known as continuous frequency distribution. There may be some variables which have integer values as well as fractional values. Frequency distribution of such variables is called continuous frequency distribution.

Let us discuss the Relative and Cumulative frequency distributions which are of the similar importance as analysis point of view of data is considered.

Relative Frequency Distribution

A relative frequency corresponding to a class is the ratio of the frequency of that class to the total frequency. The corresponding frequency distribution is called relative frequency distribution. If we multiply each relative frequency by 100, we get the percentage frequency corresponding to that class and the corresponding frequency distribution is called “Percentage frequency distribution”. Let us take an example in which both relative and percentage frequency distributions are prepared.

Example 1: A frequency distribution of marks of 50 students in a subject is as given below:

Class (Marks): 0-10 10-20 20-30 30-40 40-50

Frequency: 6 10 14 18 2

Prepare relative and percentage frequency distributions.

Solution: The relative and percentage frequency distributions can be formed as given in the following table:

Class (Marks) X	Frequency (f)	Relative frequency (f/N)	Percentage Frequency (f/N) × 100
0-10	6	$6/50 = 0.12$	$0.12 \times 100 = 12 \%$
10-20	10	$10/50 = 0.20$	$0.20 \times 100 = 20 \%$
20-30	14	$14/50 = 0.28$	$0.28 \times 100 = 28 \%$
30-40	18	$18/50 = 0.36$	$0.36 \times 100 = 36 \%$
40-50	2	$2/50 = 0.04$	$0.04 \times 100 = 4 \%$
Total	$\sum f = N = 50$	1.00	100

Cumulative Frequency Distribution

The cumulative frequency of a class is the total of all the frequencies up to and including that class. A cumulative frequency distribution is a frequency distribution which shows the observations 'less than' or 'more than' a specific value of the variable.

The number of observations less than the upper class limit of a given class is called the less than cumulative frequency and the corresponding cumulative frequency distribution is called less than cumulative frequency distribution. Similarly, the number of observations corresponding to the value of more than the lower class limit of a given class is called more than cumulative frequency and the corresponding cumulative frequency distribution is called 'more than' cumulative frequency distribution. Following is an example, wherein 'less than' and 'more than' cumulative frequency distributions have been obtained.

Example 2: For the following frequency distribution of marks of 50 students in a subject, form both types of cumulative frequency distributions.

Class (Marks)	0-10	10-20	20-30	30-40	40-50
No. of Students	7	11	15	12	5

Solution: Cumulative frequency distributions are formed as given in the following table:

Given Frequency Distribution		Less Than Cumulative Frequency Distribution		More Than Cumulative Frequency Distribution	
Classes	No. of Students	Marks Less than	No. of students	Marks More than	No of students
0-10	07	10	07	0	50
10-20	11	20	18	10	43
20-30	15	30	33	20	32
30-40	12	40	45	30	17
40-50	05	50	50	40	05
Total	50				

After discussing the frequency distributions, we now discuss how the concept of frequency distribution can be used to classify the data according to the class intervals in the next section.

2.3 GROUPING DATA IN CLASS INTERVALS

To make data understandable, data are divided into number of homogeneous groups or subgroups. In classification, according to class intervals, the observations are arranged systematically into a number of groups called classes. Such classification is most popular in practice. But before this discussion we have to define some terms which will be used in the above classification.

- (i) **Class Limits:** The class limits are the lowest and the highest values of a class. For example, let us take the class 10-20. The lowest value of this class is 10 and the highest 20. The two boundaries of a class are known as the lower limit and upper limit of the class.
- (ii) **Class Intervals:** The class interval of a class is the difference between the upper class limit and the lower class limit. For example, in the class 10-20 the class interval is 10 (i.e. 20 minus 10). This is valid in the case of exclusive method discussed in this section later. If the inclusive frequency distribution (discussed later in this section) is given then first it is converted to exclusive form and then class interval is calculated. The size of the class interval is determined by number of classes and the total range of data.
- (iii) **Range of Data:** The range of data may be defined as the difference between the lower class limit of the first class interval and the upper class limit of the last class interval.
- (iv) **Class Frequency:** The number of observations corresponding to the particular class is known as the frequency of that class or the class frequency. If we add together the frequencies of all individual classes, we obtain the total frequency.
- (v) **Class Mid Value:** It is the value lying halfway between the lower and upper class limits of class-interval, mid-point or mid value of a class is defined as follows:

$$\text{Mid Value of a Class} = \frac{\text{Upper class limit} + \text{Lower class limit}}{2}$$

For the purpose of further calculations in statistical analysis, mid value of each class is taken to represent that class.

Now we are in position to discuss the two methods of classification according to class intervals, namely “**Exclusive Method**” and “**Inclusive Method**”. Let us discuss these two methods one by one:

Exclusive Method

Under this method, a class interval is such that each upper class limit is excluded from the class interval. Here, in this method, class intervals are so fixed that the upper limit of one class is the lower limit of the next class. For example, in exclusive method a student who has secured 20 marks would be included in class 20-30, and not in 10–20. This method is widely followed in practice.

Example 3: 24 students appeared in an entrance test where all questions are objective type with 25% negative marking. The marks obtained out of 50 maximum marks are as follows:

17, 16, 7, 30, 21, 42, 44, 36, 22, 22, 25, 31, 31, 34, 30, 36, 35, 45, 25, 15, 20, 42, 40, 30

Prepare a frequency distribution by using exclusive method.

Solution: Frequency distribution of marks obtained by above 24 students is given below in Table 2.1 using the exclusive method.

Table 2.1: Frequency Distribution of 24 Students by Exclusive Method

Classes	Tally bar	No. of Students
0-10		1
10-20		3
20-30		6
30-40		9
40-50		5
Total		24

Inclusive Method

Under the inclusive method of classification both lower class limit as well as the upper limit of a class are included in that class itself. Following frequency distribution is formed using inclusive method for the data of Example 3 given above.

Table 2.2: Frequency Distribution of 24 Students by Inclusive Method

Class	Tally bar	No. of Students
0-9		1
10-19		3
20-29		6
30-39		9
40-49		5
Total		24

For converting data from *inclusive form* to *exclusive form*, first of all we find the half of the difference of lower limit of that class and upper limit of the preceding class. This value is then subtracted from lower limit of each class and added to the upper limit of each class. In the above example, this can be easily understood as $(10-9)/2 = 0.5$. So, the class intervals are as – 0.5- 9.5, 9.5-19.5, ..., 39.5-49.5. If all the observations of data are positive, then the lower limit of the first class can be taken as 0. Therefore, in this case the class intervals are as 0-9.5, 9.5-19.5, ..., 39.5-49.5.

Remark

- Lower limit of a class interval is always included in the class in both the method discussed above.
- In exclusive method upper limit of a class is not included in the class. That is why the name exclusive.

- (iii) In inclusive method upper limit of a class is also included in the class. That is why the name inclusive.

Principles of Classification

It is difficult to formulate any hard and fast rule for classifying the data. However, the following general considerations may be considered for ensuring meaningful classification of data:

- (1) The whole data should be preferably divided into number of classes between 5 and 15. To determine the approximate number of classes (K) the following formula is suggested by “Struges”:

$$K = 1 + 3.322 \text{ Log } N, \text{ where } K = \text{the approximate number of classes}$$

N = total number of observations

Log = the natural logarithm

However, the appropriate number of classes to be taken for a given data depends upon the personal judgment and other considerations such as range of data, total number of observations, etc.

- (2) One should avoid odd values of class intervals as far as possible, e.g. 3, 7, 11, 26, 39, etc. One should prefer 5 or 10 or multiple of 5 or 10 as class intervals such as 5, 10, 20, 25, 100, etc, because the human mind is accustomed more to think in terms of certain multiples of 5 or 10.
- (3) The lower class limit of the first class of a frequency distribution should either be zero or 5 or multiple of five. For example if the lowest value of the data is 26 and we have taken a class interval of 5, then the first class should be 25-30, instead of 26-31.
- (4) To maintain continuity and to get correct class interval, we should adopt exclusive method of classification. However, where ‘inclusive’ method has been adopted it is necessary to make an adjustment to determine the correct class interval and to maintain continuity.
- (5) The intervals of all the classes should be of the same size, because if the class intervals are not of the same width, it is difficult to make meaningful comparisons between classes. Sometimes the data may require the inclusion of so many class intervals that the frequency distribution will become large. Then the classification may be done as follows:

below 10; 10-20; 20-30; 30-40; above 40

These classes are called open ended classes and the distribution is known as open ended frequency distribution.

It may be noted that the frequency distributions, like other types of data presentation, are always constructed to serve some specific purpose. The technical requirements outlined above must be supplemented by sound subjective judgments if proper frequency distributions are to be formed.

Check Your Progress 1

- 1 The marks of 30 students in statistics are given below:

10, 12, 25, 32, 27, 32, 38, 43, 39, 55, 29, 38, 57, 08, 06, 13, 27, 25, 29, 53, 55, 45, 35, 48, 47, 59, 15, 19, 48, 55

Classify the above data by taking a suitable class interval.

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- 2 Present the following data of the profits (in crores of Rs.) of the 60 companies in the years 2025-26:

41, 17, 83, 63, 55, 92, 60, 58, 70, 06, 67, 82, 33, 44, 57, 49, 34, 73, 54,
63, 36, 52, 32, 75, 60, 33, 09, 79, 28, 30, 42, 93, 43, 80, 03, 32, 57, 67,
84, 64, 63, 11, 35, 28, 10, 23, 08, 41, 60, 32, 72, 53, 92, 88, 62, 55, 60,
33, 40, 57

Classify data by inclusive method.

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- 3 Use the data given in the question 2 to present the same using principle of adding and subtracting the correction factor.
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2.4 GRAPHICAL REPRESENTATION OF DATA

A graphical presentation is a geometric image of a set of data. Graphical presentation is done for both frequency distributions and time series. One of the important features of graphs is that if a person once sees the graphs, the figure representing the graphs is kept in his/her brain for a long time. They also help us in studying cause and effect relationship between two variables. The graph of a frequency distribution presents the huge data in an interesting and effective manner and brings to light the salient features of the data at a glance. Let us see some advantages of graphical presentation.

Advantages of Graphical Presentation

The following are some advantages of the graphical presentation:

- It simplifies the complexity of data and makes it readily understandable.
- It attracts attention of people.
- It saves time and efforts to understand the facts.
- It makes comparison easy.
- A graph describes the relationship between two or more variables.

Now a days a large variety of graphs are in practical use. However, we shall discuss only some important graphs which are more popularly used in practice. The various types of graphs can be divided broadly under the following two heads:

- Graphs of Frequency Distributions
- Graphs for Time Series Data

In this Unit, we will focus on graphs for frequency distributions.

2.4.1 Graphs of Frequency Distributions

The graphical presentation of frequency distributions is drawn for discrete as well as continuous frequency distributions.

Let us first consider the frequency distribution of a discrete variable.

Frequency Bar Diagram: To represent a discrete frequency distribution graphically, we take two rectangular axes of coordinates, the horizontal axis for the variable and the vertical axis for the frequency. The different values of the variable are then located as points on the horizontal axis. At each of these points, a perpendicular bar is drawn to present the corresponding frequency.

Such a diagram is called a 'Frequency Bar Diagram'. For example, if we take the frequency distribution for the number of peas per pod for 198 pods as given in Table 2.3:

Table 2.3: A Sample Frequency Distribution

No of peas per pod	1	2	3	4	5	6	7
Frequency (number of pods)	14	23	66	40	26	18	11

Then, the frequency bar diagram is shown in Fig. 2.1:

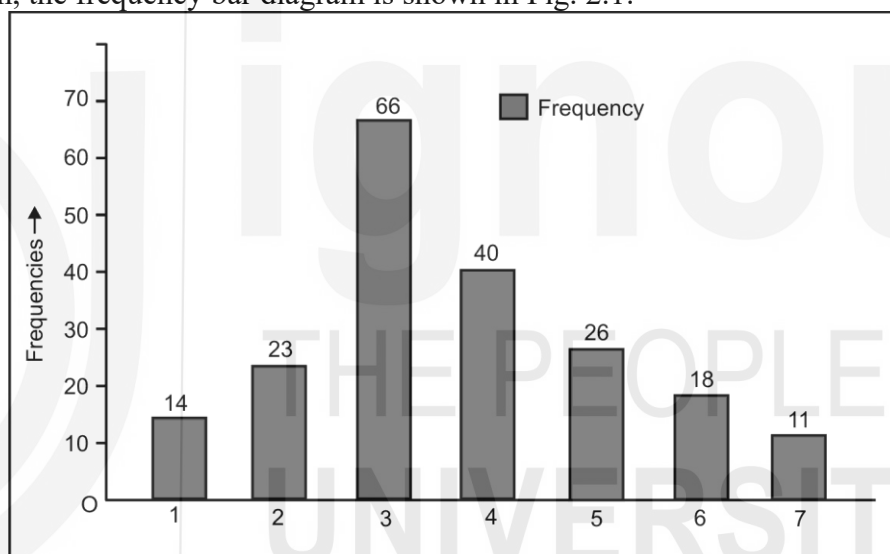


Fig. 2.1: Frequency Bar Diagram for the Frequency Distribution of Number of the Peas for 198 Pods.

Note 1: O represents origin and choice of scale used along horizontal and vertical axes depends upon given data.

Now, we take the case of frequency distribution of a continuous variable.

The following are the most commonly used graphs for continuous frequency distributions:

- (i) Histogram
 - (ii) Frequency Polygon
 - (iii) Frequency Curve
 - (iv) Cumulative Frequency Curve or Ogives
- Let us discuss these one by one:

(i) Histogram

In previous example, we have discussed how a graph is drawn for discrete frequency distribution.

For the continuous frequency distribution, a better way to represent the data graphically is to use a histogram. A histogram is drawn by constructing adjacent rectangles over the class intervals such that the length of the rectangles is proportional to the corresponding class frequencies.

Histogram is similar to a bar diagram which represents a frequency distribution with continuous classes. The width of all bars is equal to class interval. Each rectangle is joined with the other so as to give a continuous picture.

The class-boundaries are located on the horizontal axis. If the class-intervals are of equal size, the heights of the rectangles will be proportional to the class-frequencies themselves. If the class-intervals are not of equal size, the heights of the rectangles will be proportional to the ratios of the frequencies to the width of the corresponding classes. In other words, the frequencies of the class-intervals having the least width are written as they are and the frequencies of other class intervals are written as follows:

$$\frac{\text{Given frequency}}{\text{Width of its Class-interval}} \times (\text{The least width}) \quad \dots (1)$$

Let us draw a histogram to the following frequency distribution given below in the Table 2.4.

Table 2.4: A Sample Frequency Distribution for Class Intervals of Equal Width

Class Intervals	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	2	3	13	18	9	7	6	2

Histogram for the above data is given below.

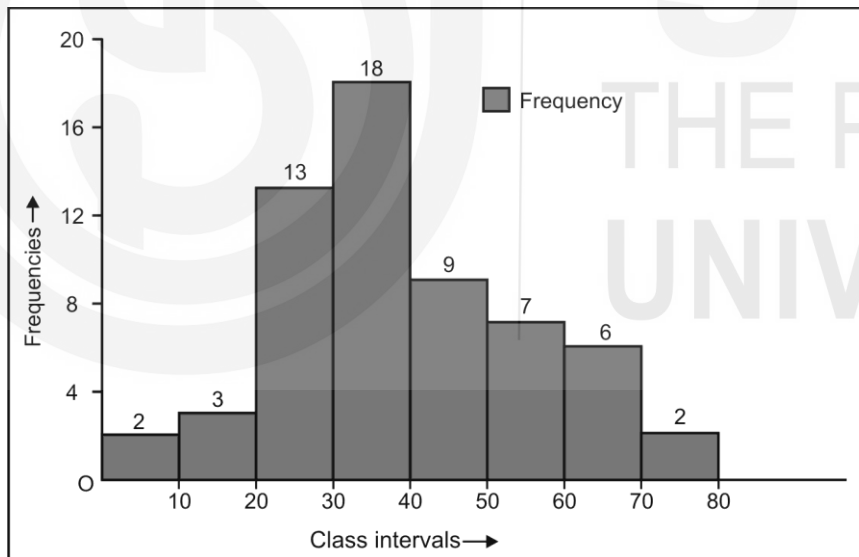


Fig. 2.2: Histogram for Frequency Distribution when Class-intervals are of Equal Width.

Now, let us consider the frequency distribution for unequal class intervals as given in Table 2.5

Table 2.5: A Frequency Distribution for Class Intervals of Unequal Width

Class	0-10	10-20	20-30	30-40	40-70	70-80	80-100
Frequency	20	32	8	2	60	35	10

As it is a case of unequal class intervals, so we have to adjust the frequencies of the classes 40-70 and 80-100 by the formula suggested in equation 1. These calculations are shown in Table 2.6 given below:

Table 2.6: Computation of Height of Rectangles in Histograms

Class Interval (CI)	Frequency	Width of (CI)	Heights of the rectangles
0-10	20	10	20
10-20	32	10	32
20-30	8	10	8
30-40	2	10	2
40-70	60	30	$(60/30) \times 10 = 20$
70-80	35	10	35
80-100	10	20	$(10/20) \times 10 = 5$

The histogram for this frequency distribution is shown in Fig. 2.3.

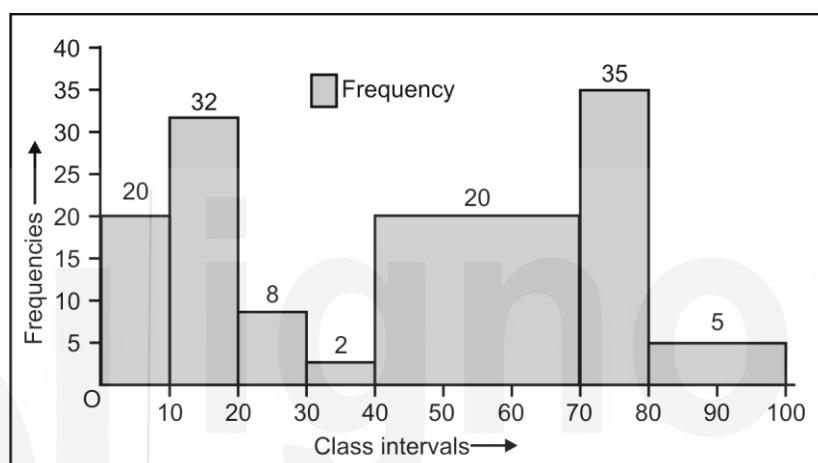


Fig. 2.3: Histogram for Frequency Distribution when Class Intervals are of Unequal Width

Note 2: Sometimes, a histogram is also used for the frequency distribution of a discrete variable. Each value of the discrete variable is regarded as the mid-point of an interval. But generally, its use is not recommended, because in discrete case each frequency actually corresponds to a single point and not to an interval.

(ii) Frequency Polygon

Another method of presenting a frequency distribution graphically other than histogram is to use a frequency polygon. In order to draw the graph of a frequency polygon, first of all the mid values of all the class intervals and the corresponding frequencies are plotted as points with the help of the rectangular coordinate axes. Secondly, we join these plotted points by line segments. The graph thus obtained is known as frequency polygon, but one important point to keep in mind is that whenever a frequency polygon is required, we take two imaginary class intervals each with frequency zero, one just before the first class interval and other just after the last class interval. Addition of these two class intervals facilitate the existence of the property that

$$\text{Area under the polygon} = \text{Area of the histogram}$$

For example, if we take the frequency distribution as given in Table 2.4, then we have to first plot the points (5, 2), (15, 3), ..., (75, 2) on graph paper along with the horizontal bars. Then we join the successive points (including the midpoints of two imaginary class intervals, each with zero frequency) by line segments to

get a frequency polygon. The frequency polygon for frequency distribution given in Table 2.4 is shown in Fig. 2.4.

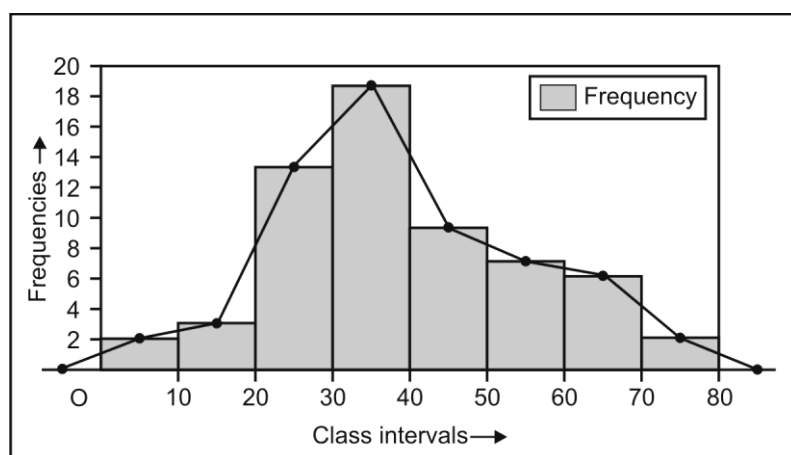


Fig. 2.4: Frequency Polygon for the Frequency Distribution given in Table 2.4.

Note 3: In some cases, the first class interval does not start from zero. In such situations, we mark a kink on the horizontal axis, which will indicate the continuity of the scale starting from zero. Let us take an example of this type.

(iii) Frequency Curve

In simple words frequency curve is a smooth curve obtained by joining the points (not necessarily all points) of the frequency polygon such that

- Like frequency curve, it also starts from the baseline (horizontal axis) and ends at the baseline.
- Area under frequency curve remains approximately equal to the area under the frequency polygon.

In other words, let us try to explain the concept theoretically. Suppose we draw a sample of size n from a large population. Frequency curve is the graph of a continuous variable. So, theoretically, continuity of the variable implies that whatever small class interval we take, there will be some observations in that class interval. That is, in this case, there will be a large number of line segments and the frequency polygon tends to coincide with the smooth curve passing through these points as the sample size (n) increases. This smooth curve is known as frequency curve.

In the following example, we have drawn both frequency polygon and frequency curve to make the idea clear for you.

Example 4: Draw frequency polygon and frequency curve for the following frequency distribution.

Class Intervals	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
frequency	2	5	8	15	18	10	3	1

Solution: Frequency polygon and frequency curve for the above data is given below in Fig. 2.5.

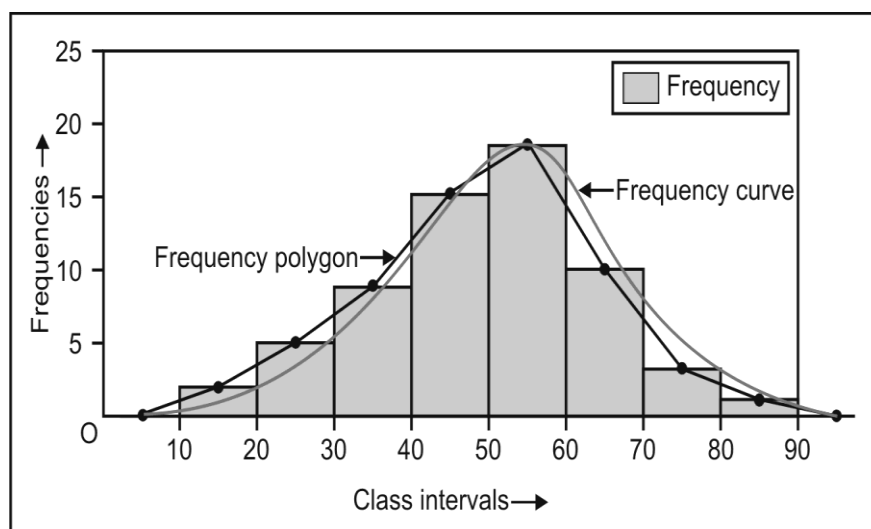


Fig. 2.5: Frequency Curve along with Frequency Polygon.

On the next page some important types of frequency curves are given which are generally obtained in the graphical presentations of frequency distributions. That is, symmetrical, positively skewed, negatively skewed, J shaped, U shaped, bimodal and multimodal frequency curves. You note that the shapes of these curves justify their names.

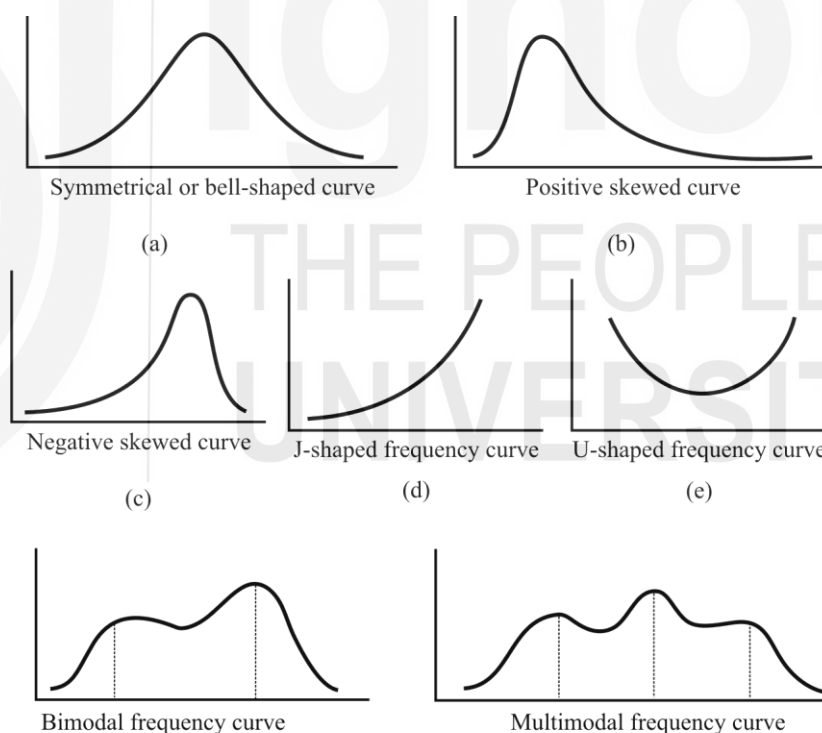


Fig. 2.6: Some Important Shapes of Curves

(iv) Cumulative Frequency Curves - Ogive

The concept of cumulative frequency distribution involves less than cumulative frequency distribution or more than cumulative frequency distribution. Here our aim is to study graphical presentation of less than and more than cumulative frequency distributions, which are known as less than frequency curve (or less than ogive) and more than frequency curve (or more than ogive) respectively.

For drawing less than cumulative frequency curve (or less than ogive), first of all the cumulative frequencies are plotted against the values (upper limits of the class intervals) up to which they correspond and then we simply join the points by line segments, curve thus obtained is known as less than ogive. Similarly, more than frequency curve (more than ogive) can be obtained by plotting more than cumulative frequencies against lower limits of the class intervals. As we have already mentioned within brackets that less than cumulative frequency curve and more than cumulative frequency curve are also called **less than ogive** and **more than ogive** respectively.

In other words we may define less than ogive and more than ogive as follow:

Less Than Ogive: If we plot the points with the upper limits of the classes as abscissae and the cumulative frequencies corresponding to the values less than the upper limits as ordinates and join the points so plotted by line segments, the curve thus obtained is nothing but known as “less than cumulative frequency curve” or “less than ogive”. It is a rising curve.

More Than Ogive: If we plot the points with the lower limits of the classes as abscissae and the cumulative frequencies corresponding to the values more than the lower limits as ordinates and join the points so plotted by line segments, the curve thus obtained is nothing but known as “more than cumulative frequency curve” or “more than ogive”. It is a falling curve.

Let us draw both the ogives (‘less than’ and ‘more than’) for the following frequency distribution of the weekly wages of the number of workers given in Table 2.7.

Table 2.7: A Sample Frequency Distribution

Weekly wages	0-10	10-20	20-30	30-40	40-50
No. of workers	45	55	70	40	10

Before drawing the ogives, we make a cumulative frequency distribution as given in Table 2.8

Table 2.8: Cumulative Frequency Distribution of Data of Table 2.7

Weekly wages	No. of workers	Less than Cumulative frequency distribution		More than Cumulative frequency distribution	
		Wages Less than	Number of workers	Wages More than	Number of workers
0-10	45	10	45	0	220
10-20	55	20	100	10	175
20-30	70	30	170	20	120
30-40	40	40	210	30	50
40-50	10	50	220	40	10

From the above data, we construct both the ogives as shown in Fig. 2.7 and Fig. 2.8. For “less than ogive” as shown on previous page in Fig. 2.7, we have plotted the points (10, 45), (20, 100), (30, 170), (40, 210), (50, 220) and then joined them by line segments. Similarly, for “more than ogive” as shown above in Fig. 2.8, we have plotted the points (0, 220), (10, 175), (20, 120), (30, 50), (40, 10), and then joined them by line segments.

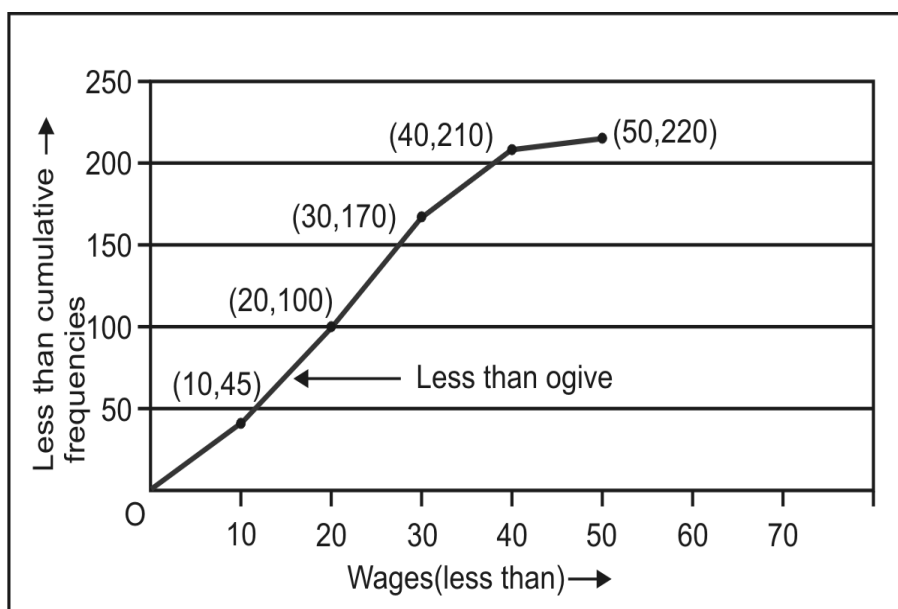


Fig. 2.7: Less Than Ogive.

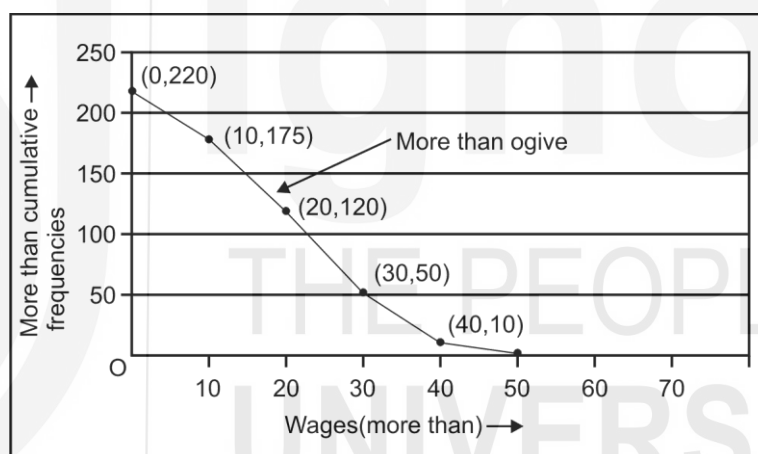


Fig. 2.8: More Than Ogive.

If we want to obtain a partition value, using ogives, we draw dotted horizontal line through that value at y-axis which corresponds to the partition value and then from the point, where it meets the less than ogive, we draw a dotted vertical line and let it meet the x-axis. The abscissa of the point, where it meets the x-axis is the required partition value. For example, suppose we want to find first quartile, then we draw a dotted horizontal line starting from y-axis at a point corresponding to $N/4$ and let it meet the “less than ogive”. From that point at “less than ogive”, we draw a dotted vertical line and let it meet the x-axis. The abscissa corresponding to this point is the first quartile. Similarly, for finding median or second quartile, we start drawing dotted horizontal line from y-axis at a point corresponding to $N/2$ and then we proceed as described above. Similarly, for third quartile $3N/4$ is taken in place of $N/2$. In this way, we may find any partition value.

Note 4: Median may also be obtained by drawing dotted vertical line through the point of intersection of both the ogives, when drawn on a single figure.

2.4.2 Stem-And-Leaf Plots

A stem-and-leaf display is very similar to a histogram but shows more information. The stem-and-leaf display summarises the shape of a set of data and provides the details regarding individual values. A stem-and-leaf display quickly summarises data while maintaining the individual data points. Now a day's use of stem-and-leaf displays is increasing, so let us formally define it in the next paragraph with some examples.

It has a vertical line of numbers obtained after removing the last digits (i.e. unit digits) from the given numbers called starting parts and for each starting part there is a horizontal line of numbers, i.e. the digits at the unit places of the given numbers called leaves. And each complete horizontal line including starting part and leaves is known as stem. The data displayed like this is nothing but known as stem-and-leaf display. The distance between the lowest values that are recorded in two consecutive stems is known as **stem width** or **category interval**, which plays very important role in stem-and-leaf displays.

Stem-and-leaf displays are used in many situations like series of scores of sports teams, series of temperature or rainfall over a period of time, series of classroom test scores, etc.

The following example will illustrate the above discussion more clearly.

Example 5: We have a set of values of the test scores of 22 students in a class as 11, 2, 28, 33, 48, 0, 42, 17, 24, 14, 0, 18, 26, 29, 35, 42, 22, 8, 28, 8, 46, 14. Draw a simple stem-and-leaf display by taking stem width 10.

Solution: Simple stem-and-leaf display for the given data can be shown as follows:

0		2 0 0 8 8	(5)
1		1 7 4 8 4	(5)
2		8 4 6 9 2 8	(6)
3		3 5	(2)
4		8 2 2 6	(4)

Total observations = 22

After arranging the leaves in ascending order of magnitude, we have

0		0 0 2 8 8
1		1 4 4 7 8
2		2 4 6 8 8 9
3		3 5
4		2 2 6 8

Here starting parts show the 'tens digits' and the leaves show the 'ones digits' in the above stem-and-leaf display. At a glance, one can see that 4 students got marks in the 40's in their test out of 50. Out of these four students two got 42 marks each, whereas the other two got 46 and 48 marks in the test. Fourth row (i.e. fourth stem) indicates that two students got marks in 30's in their test out of 50. And actual marks of these two students are 33 and 35. Similarly, we can get the information about the marks of the other students from successive rows (stems). When you count the total numbers of leaves, you may know how many students appeared in the test. The information is nicely organised when a stem-and-leaf display is used. Stem-and-leaf display provides a tool for specific

information in large sets of data, otherwise one would have a long list of marks to arrange and analyse.

Stem-and-Leaf Display for more than one Set of Data

Stem-and-leaf display is also used to compare two sets of data. That is known as 'back to back' stem-and-leaf display. For example, if you want to compare the batting scores of two cricket players, then stem-and-leaf display is right way to represent the data.

Example 6: Draw a stem-and-leaf display for batting scores of two players given below.

Player A	102, 61, 82, 88, 90, 63, 69, 85, 105, 93, 65, 94, 107, 97, 67
Player B	104, 62, 83, 95, 106, 95, 108, 63, 108, 82, 93, 109

Solution: The scores of two players can be compared with the help of back to back stem-and-leaf display as follows:

Leaf (Player A)	Starting part	Leaf (Player B)
1 3 5 7 9	6	2 3
2 5 8	8	2 3
0 3 4 7	9	3 5 5
2 5 7	10	4 6 8 8 9

Here column of starting parts is now in the middle and the leaves columns are to the right (player B) and left (player A) of the column of starting parts. You can see that the player B has more innings with a highest score than the player A. The player B has only 2 innings with scores of 62 and 63, while the player A has 5 innings with the scores of 61, 63, 65, 67 and 69. You can also see that player B has the highest score of 109, compared to player A with highest score of 107. Thus we see that presentation of the data by stem-and-leaf display provides us lot of information in very quick time.

Note: Stem-and-leaf display arranges the data in place values. They show the shape of data and can help to find median, mode and quartiles of data.

Check Your Progress 2

- Draw a histogram for the following frequency distribution
Wage (Rs): 0-5 5-10 10-15 15-20 20-25 25-30 30-35 35-40 40-45 45-50
No. of
Workers: 30 70 100 110 140 150 130 100 90 60
- Draw a frequency polygon from the following data
Wage: 0-5 5-10 10-15 15-20 20-25 25-30 30-35 35-40 40-45 45-50
No of
Workers: 20 50 80 110 140 150 120 150 100 80
- Draw a frequency curve from the following frequency distribution
Class Intervals: 1-2 2-3 3-4 4-5 5-6 6-7 7-8 8-9
Frequency: 3 9 20 25 18 12 6 3
- Draw two ogives from the following data
Class: 0-10 10-20 20-30 30-40 40-50 50-60 60-70 70-80 80-90
Frequency: 3 6 10 13 20 18 15 9
- Draw a stem-and-leaf display with the following marks obtained by 30 students.
77, 80, 82, 68, 65, 59, 61, 57, 50, 62, 61, 70, 69, 64, 67,
70, 62, 65, 65, 73, 76, 87, 80, 82, 83, 79, 79, 71, 80, 77

2.5 MEASURES OF CENTRAL TENDENCY

According to **Professor Bowley**, averages are “statistical constants which enable us to comprehend in a single effort the significance of the whole”. They throw light as to how the values are concentrated in the central part of the distribution. For this reason they are also called the measures of central tendency, an average is a single value which is considered as the most representative for a given set of data. Measures of central tendency show the tendency of some central value around which data tend to cluster.

Why use the Measure of Central Tendency?

The following are two main reasons for studying an average:

1. **To get a single representative:** A Measure of central tendency enables us to get a single value from the mass of data and also provides an idea about the entire data. For example, it is impossible to remember the height measurements of all students in a class. But if the average height is obtained, we get a single value that represents the entire class.
2. **To facilitate comparison:** Measures of central tendency enable us to compare two or more populations by reducing the mass of data in a single figure. The comparison can be made either at the same time or over a period of time. For example, if a subject has been taught in more than two classes, the average marks of those classes can be obtained, and a comparison can be made.

What should be the Properties of a Good Average?

The following are the properties of a good measure of average:

1. It should be simple to understand, as its purpose is to simplify complexity.
2. It should be easy to calculate so that it can be used as widely as possible.
3. It should be defined unambiguously so that if different people compute the average from same figures, they get the same answer.
4. It should be liable for algebraic manipulations. For example, if there are two sets of data and the individual information is available for both set, then one can be able to find the information regarding the combined set also then something is missing.
5. It should be least affected by sampling fluctuations. In other words, if we select 10 different groups of observations from same population and compute the average of each group, then we should expect to get approximately the same values. There may be little difference because of the sampling fluctuation only.
6. It should be based on all the observations, that is each and every observation is used for its calculation.
7. It should be possible to calculate even for open-end class intervals
8. It should not be affected by extremely small or extremely large observations. If one or two very small or very large observations affect the average i.e. either increase or decrease its value largely, then the average cannot be considered as a good average.

Different Measures of Central Tendency

The following are the various measures of central tendency:

1. Arithmetic Mean
2. Weighted Mean
3. Median
4. Mode
5. Geometric Mean
6. Harmonic Mean

The next sub-sections discuss them in more detail

2.5.1 Mean

Arithmetic mean (also called mean) is defined as the sum of all the observations divided by the number of observations. Arithmetic mean (AM) may be calculated for the following two types of data:

1. For Ungrouped Data

For ungrouped data, arithmetic mean may be computed by applying any of the following methods:

(1) Direct Method

Mathematically, if $x_1, x_2, \dots, x_n$ are the n observations then their mean is

$$\bar{X} = \frac{(x_1 + x_2 + x_3 + \dots + x_n)}{n}$$

$$\bar{X} = \frac{\sum_{i=1}^n x_i}{n}$$

If f_i is the frequency of x_i ($i=1, 2, \dots, k$), the formula for arithmetic mean would be

$$\bar{X} = \frac{(f_1 x_1 + f_2 x_2 + \dots + f_k x_k)}{(f_1 + f_2 + \dots + f_k)}$$

$$\bar{X} = \frac{\sum_{i=1}^k f_i x_i}{\sum_{i=1}^k f_i}$$

(2) Short-cut Method

The arithmetic mean can also be calculated by taking deviations from any arbitrary point “A”, in which the formula shall be

$$\bar{X} = A + \frac{\sum_{i=1}^n d_i}{n} \quad \text{where, } d_i = x_i - A$$

If f_i is the frequency of x_i ($i=1, 2, \dots, k$), the formula for arithmetic mean would be

$$\bar{X} = A + \frac{\sum_{i=1}^k f_i d_i}{\sum_{i=1}^k f_i}, \quad \text{where, } d_i = x_i - A$$

Here, k is the number of distinct observations in the distribution.

Note: Usually, the shortcut method is used when data values are large.

Example 7: Calculate mean of the weights of five students by using shortcut method.

54, 56, 70, 45, 50 (in kg)

Solution: For shortcut method, we use following formula

$$\bar{X} = A + \frac{\sum d_i}{n}, \quad \text{where } d_i = x_i - A$$

If 50 is taken as the assumed value A in the given data in then, for the calculation of d_i we prepare following table:

We have $A = 50$ then,

$$\begin{aligned} \bar{X} &= A + \frac{\sum_{i=1}^n d_i}{n} \\ &= 50 + \frac{25}{5} = 50 + 5 = 55 \end{aligned}$$

x	$d = x - A$
54	$54 - 50 = 4$
56	$56 - 50 = 6$
70	$70 - 50 = 20$
45	$45 - 50 = -5$
50	$50 - 50 = 0$
	$\sum_{i=1}^n d_i = 25$

2 For Grouped Data

Direct Method

If f_i is the frequency of x_i ($i = 1, 2, \dots, k$) where x_i is the mid value of the i^{th} class interval, the formula for arithmetic mean would be

$$\bar{X} = \frac{(f_1 x_1 + f_2 x_2 + \dots + f_k x_k)}{(f_1 + f_2 + \dots + f_k)}$$

$$\bar{X} = \frac{\sum_{i=1}^k f_i x_i}{\sum_{i=1}^k f_i} = \frac{\sum fx}{\sum f} = \frac{\sum fx}{N},$$

where, $N = f_1 + f_2 + \dots + f_k$

Short-cut Method

$$\bar{X} = A + \frac{\sum_{i=1}^k f_i d_i}{\sum_{i=1}^k f_i}, \quad \text{where, } d_i = x_i - A$$

Here, f_i would be representing the frequency of the i^{th} class, x_i is the mid-value of the i^{th} class and k is the number of classes.

Example 8: For the following data, calculate arithmetic mean using direct method:

Class Interval	0-10	10-20	20-30	30-40	40-50
Frequency	3	5	7	9	4

Solution: We have the following distribution:

Class Interval	Mid Value x	Frequency f	fx
0-10	05	03	15
10-20	15	05	75
20-30	25	07	175
30-40	35	09	315
40-50	45	04	180
		$\sum_{i=1}^k f_i = N = 28$	$\sum_{i=1}^k f_i x_i = 760$

$$\text{Mean} = \frac{\sum_{i=1}^k f_i x_i}{N} = 760/28 = 27.143$$

Note on Arithmetic Mean

Arithmetic mean fulfills most of the properties of a good average except the last two. It is particularly useful when we are dealing with a sample as it is least affected by sampling fluctuations. It is the most popular average and should always be our first choice unless there is a strong reason for not using it.

2.5.2 Median

Median is that value of the variable which divides the whole distribution into two equal parts. Here, it may be noted that the data should be arranged in ascending or descending order of magnitude. When the number of observations is odd then the median is the middle value of the data. For even number of observations, there will be two middle values. So we take the arithmetic mean of these two middle values. Number of the observations below and above the median, are same. Median is not affected by extremely large or extremely small values (as it corresponds to the middle value) and it is also not affected by open end class intervals. In such situations, it is preferable in comparison to mean. It is also useful when the distribution is skewed (asymmetric). Skewness will be discussed in the next unit.

1. Median for Ungrouped Data

Mathematically, if $x_1, x_2, \dots, x_n$ are the n observations then for obtaining the median first of all we have to arrange these n values either in ascending order or in descending order. When the observations are arranged in ascending or descending order, the middle value gives the median if n is odd. For even number of observations there will be two middle values. So we take the arithmetic mean of these two values.

$$M_d = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ observation ; (when } n \text{ is odd)}$$

$$M_d = \frac{\left(\frac{n}{2}\right)^{\text{th}} \text{ observation} + \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ observation}}{2}; \text{ (when n is even)}$$

For Ungrouped Data (when frequencies are given)

If x_i are the different value of variable with frequencies f_i then we calculate cumulative frequencies from f_i then median is defined by

$$M_d = \text{Value of variable corresponding to } \left(\frac{\sum f}{2}\right)^{\text{th}}$$

$$= \left(\frac{N}{2}\right)^{\text{th}} \text{ cumulative frequency.}$$

Note: If $N/2$ is not the exact cumulative frequency then value of the variable corresponding to next cumulative frequencies is the median.

Example 9: Find Median from the given frequency distribution

x	20	40	60	80
f	7	5	4	3

Solution: First we find cumulative frequency

x	f	c.f.
20	7	7
40	5	12
60	4	16
80	3	19
	$\sum_{i=1}^k f_i = 19$	

$$M_d = \text{Value of the variable corresponding to the } \left(\frac{19}{2}\right)^{\text{th}} \text{ cumulative frequency}$$

$$= \text{Value of the variable corresponding to 9.5 since 9.5 is not among c.f.}$$

So, the next cumulative frequency is 12 and the value of variable against 12 cumulative frequency is 40. So median is 40.

2. Median for Grouped Data

For class interval, first we find cumulative frequencies from the given frequencies and use the following formula for calculating the median:

$$\text{Median} = L + \frac{\left(\frac{N}{2} - C\right)}{f} \times h$$

where, L = lower class limit of the median class,

N = total frequency,

C = cumulative frequency of the pre-median class,

f = frequency of the median class, and

h = width of the median class.

Median class is the class in which the $(N/2)^{\text{th}}$ observation falls. If $N/2$ is not among any cumulative frequency then next class to the $N/2$ will be considered as median class.

Example 10: Calculate median for the data given in Example 8.

Solution: We first form a cumulative frequency table (cumulative frequency of a class gives the number of observations less than the upper limit of the class; strictly speaking, this is called cumulative frequency of less than type; we also have cumulative frequency of more than type which gives the number of observations greater than or equal to the lower limit of a class):

Class Interval	Frequency f	Cumulative Frequency (< type)
0-10	3	3
10-20	5	8
20-30	7	15
30-40	9	24
40-50	4	28

$$\sum_{i=1}^k f_i = N = 28 \quad \Rightarrow \quad \frac{N}{2} = \frac{28}{2} = 14$$

Since 14 is not among the cumulative frequency so the class with next cumulative frequency i.e. 15, which is 20-30, is the median class.

We have

L = lower class limit of the median class = 20

N = total frequency = 28

C = cumulative frequency of the pre median class = 8

f = frequency of the median class = 7

h = width of median class = 10

Now substituting all these values in the formula of Median

$$\text{Median} = L + \frac{\left(\frac{N}{2} - C\right)}{f} \times h$$

$$M_d = 20 + \frac{14 - 8}{7} \times 10 = 28.57$$

Therefore, median is 28.57.

Note: Median is easy to understand and compute, and is not affected by extremely small or extremely large values; however, it does not utilise all the observations. For example, the median of 1, 2, 3 is 2. If observation 3 is replaced by any number higher than or equal to 2 and if the number 1 is replaced by any number lower than or equal to 2, the median value will be unaffected. This means 1 and 3 are not being utilised.

2.5.3 Mode

Highest frequent observation in the distribution is known as mode. In other words, mode is that observation in a distribution which has the maximum frequency. For example, when we say that the average size of shoes sold in a shop is 7 it is the modal size which is sold most frequently.

For Ungrouped Data

Mathematically, if $x_1, x_2, \dots, x_n$ are the n observations and if some of the observation are repeated in the data, say x_i is repeated highest times then we can say the x_i would be the mode value.

For Grouped Data:

Data where several classes are given, following formula of the mode is used

$$M_0 = L + \frac{|f_1 - f_0|}{|f_1 - f_0| + |f_1 - f_2|} \times h$$

where, L = lower class limit of the modal class,

f_1 = frequency of the modal class,

f_0 = frequency of the pre-modal class,

f_2 = frequency of the post-modal class, and

h = width of the modal class.

Modal class is that class which has the maximum frequency.

Example 11: For the data given in Example 8, calculate mode.

Solution: Here the frequency distribution is

Class Interval	Frequency
0-10	3
10-20	5
30-40	7
40-50	9
50-60	4

Corresponding to highest frequency 9 modal class is 40-50 and we have

$$L = 40, f_1 = 9, f_0 = 7, f_2 = 4, h = 10$$

Applying the formula,

$$\begin{aligned} \text{Mode} &= 40 + \frac{(9 - 7)}{(2 \times 9 - 7 - 4)} \times 10 \\ &= 42.86 \end{aligned}$$

Note: Mode is the easiest average to understand and also easy to calculate and is not affected by extreme values. Mode can be calculated even if the classes are of unequal width. However, a distribution can have more than one mode, and it does not utilize all the observations. Further, Mode is not amenable to algebraic treatment; an is greatly affected by sampling fluctuations.

2.5.4 Relationship between Mean, Median and Mode

For a symmetrical distribution the mean, median and mode coincide. But if the distribution is moderately asymmetrical, there is an empirical relationship between them. The relationship is

$$\text{Mean} - \text{Mode} = 3 (\text{Mean} - \text{Median})$$

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

Note: Using this formula, we can calculate mean/median/mode if other two of them are known.

2.5.5 Geometric and Harmonic Mean

Geometric Mean: The geometric mean (GM) of n observations is defined as the n -th root of the product of the n observations. It is useful for averaging ratios or proportions. It is the ideal average for calculating index numbers (index numbers are economic barometers which reflect the change in prices or commodity consumption in the current period with respect to some base period taken as standard). It fails to give the correct average if an observation is zero or negative.

1. For Ungrouped Data

If $x_1, x_2, \dots, x_n$ are the n observations of a variable X then their geometric mean is

$$GM = \sqrt[n]{x_1 x_2 \dots x_n}$$

$$GM = (x_1 x_2 \dots x_n)^{\frac{1}{n}}$$

Taking log of both sides

$$\log GM = \frac{1}{n} \log (x_1 x_2 \dots x_n)$$

$$\log GM = \frac{1}{n} (\log x_1 + \log x_2 + \dots + \log x_n)$$

$$\Rightarrow GM = \text{Antilog} \left(\frac{1}{n} \sum_{i=1}^n \log x_i \right)$$

Example 12: Find geometric mean of 2, 4, 8.

Solution: $GM = (x_1 x_2 \dots x_n)^{\frac{1}{n}}$

$$GM = (2 \times 4 \times 8)^{\frac{1}{3}} = (64)^{\frac{1}{3}} = 4$$

Thus, the geometric mean is 4.

2. For Grouped data

If $x_1, x_2, \dots, x_k$ are k values (or mid values in case of class intervals) of a variable X with frequencies $f_1, f_2, \dots, f_k$ then

$$GM = (x_1^{f_1} x_2^{f_2} \dots x_k^{f_k})^{\frac{1}{\sum f_i}}$$

$$GM = (x_1^{f_1} x_2^{f_2} \dots x_k^{f_k})^{\frac{1}{N}}$$

where $N = f_1 + f_2 + \dots + f_k$

Taking log of both sides

$$\log GM = \frac{1}{N} \log (x_1^{f_1} x_2^{f_2} \dots x_k^{f_k})$$

$$\log GM = \frac{1}{N} (f_1 \log x_1 + f_2 \log x_2 + \dots + f_k \log x_k)$$

$$\Rightarrow GM = \text{Antilog} \left(\frac{1}{N} \sum_{i=1}^k f_i \log x_i \right)$$

Example 13: For the data in Example 8, calculate geometric mean.

Solution:

Class	Mid Value (x)	Frequency (f)	log x	f × log x
0-10	5	3	0.6990	2.0970
10-20	15	5	1.1761	5.8805
20-30	25	7	1.3979	9.7853
30-40	35	9	1.5441	13.8969
40-50	45	4	1.6532	6.6128
		N = 28		$\sum f_i \log x_i = 38.2725$

Using the formula

$$\begin{aligned}
 \text{GM} &= \text{Antilog} \left(\frac{1}{N} \sum_{i=1}^k f_i \log x_i \right) \\
 &= \text{Antilog} \left(\frac{38.2725}{28} \right) \\
 &= \text{Antilog} (1.3669) \\
 &= 23.28
 \end{aligned}$$

Note: Geometric mean uses all the observations and gives more weight to small items. It is not affected greatly by sampling fluctuations. However, it is difficult calculate; and becomes imaginary for an odd number of negative observations and becomes zero or undefined if a single observation is zero.

Harmonic Mean: The harmonic mean (HM) is defined as the reciprocal (inverse) of the arithmetic mean of the reciprocals of the observations of a set.

1. For Ungrouped Data

If $x_1, x_2, \dots, x_n$ are the n observations of a variable X , then their harmonic mean is

$$\begin{aligned}
 \text{HM} &= \frac{1}{\frac{1}{n} \left[\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} \right]} \\
 \text{HM} &= \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}
 \end{aligned}$$

Example 14: Calculate the Harmonic mean of 1, 3, 5, 7

Solution: Formula for harmonic mean is

$$\text{HM} = \frac{1}{\frac{1}{n} \left[\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} \right]}$$

$$\begin{aligned}
 HM &= \frac{1}{\frac{1}{4} \left[\frac{1}{1} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} \right]} \\
 &= 4 / (1.0000 + 0.3333 + 0.2000 + 0.1428) \\
 &= 4/1.6761 = 2.39
 \end{aligned}$$

2. For Grouped Data

If $x_1, x_2, \dots, x_k$ are k values (or mid values in case of class intervals) of a variable X with their corresponding frequencies $f_1, f_2, \dots, f_k$, then

$$\begin{aligned}
 HM &= \frac{1}{\frac{1}{N} \left[\frac{f_1}{x_1} + \frac{f_2}{x_2} + \dots + \frac{f_k}{x_k} \right]} \\
 HM &= \frac{N}{\sum_{i=1}^k \frac{f_i}{x_i}} \quad \text{where, } N = \sum_{i=1}^k f_i
 \end{aligned}$$

When equal distances are travelled at different speeds, the average speed is calculated by the harmonic mean. It cannot be calculated if an observation is zero.

Example 15: For the data given in Example 8, calculate the harmonic mean.

Solution:

Class	Mid Value (x)	Frequency (f)	f/x
0-10	5	3	0.600
10-20	15	5	0.330
20-30	25	7	0.280
30-40	35	9	0.257
40-50	45	4	0.088
		$N = \sum f = 28$	$\sum f/x = 1.555$

Using the formula,

$$\begin{aligned}
 HM &= \frac{N}{\sum_{i=1}^k \frac{f_i}{x_i}} \\
 &= 28/1.555 = 17.956
 \end{aligned}$$

Note: Harmonic mean also uses all the observations, and gives greater importance to small items; however, it is difficult to understand and compute.

Relations between AM, GM and HM

We state the relationships without any proof. You may refer to the references for proof.

Relation 1: $AM \geq GM \geq HM$

Relation 2: $GM = \sqrt{AM \cdot HM}$

Check Your Progress 3

- 1 Find arithmetic mean of the distribution of marks given below:

Marks	0-10	10-20	20-30	30-40	40-50
No. of students	6	9	17	10	8

2. Find the missing frequency when median is given as Rs 50.

Expenditure (Rs)	0-20	20-40	40-60	60-80	80-100
No. of families	5	15	30	--	8

- 3 In an asymmetrical distribution the mode and mean are 35.4 and 38.6 respectively. Calculate the median.

.....

2.6 PARTITION VALUES

Partition values are those values of variable which divide the distribution into a certain number of equal parts. Here it may be noted that the data should be arranged in ascending or descending order of magnitude. Commonly used partition values are quartiles, deciles and percentiles. For example, quartiles divide the data into four equal parts. Similarly, deciles and percentiles divide the distribution into ten and hundred equal parts, respectively.

2.6.1 Quartiles

Quartiles divide whole distribution into four equal parts. There are three quartiles- 1st Quartile denoted as Q_1 , 2nd Quartile denoted as Q_2 and 3rd Quartile as Q_3 , which divide the whole data in four parts. 1st Quartile contains the $\frac{1}{4}$ part of data, 2nd Quartile contains $\frac{1}{2}$ of the data and 3rd Quartile contains the $\frac{3}{4}$ part of data. Here, it may be noted that the data should be arranged in ascending or descending order of magnitude.

For Ungrouped Data

For obtaining the quartiles first of all we have to arrange the data either in ascending order or in descending order of their magnitude. Then, we find the $(N/4)^{\text{th}}$, $(N/2)^{\text{th}}$ and $(3N/4)^{\text{th}}$ placed item in the arranged data for finding out the 1st Quartile (Q_1), 2nd Quartile (Q_2) and 3rd Quartile (Q_3) respectively. The value of the $(N/4)^{\text{th}}$ placed item would be the 1st Quartile (Q_1), value of $(N/2)^{\text{th}}$ placed item would be the 2nd Quartile (Q_2) and value of $(3N/4)^{\text{th}}$ placed item would be the 3rd Quartile (Q_3) of that data.

Mathematically, if $x_1, x_2, \dots, x_N$ are N values of a variable X then 1st Quartile (Q_1), 2nd Quartile (Q_2) and 3rd Quartile (Q_3) are defined as:

First Quartile (Q_1) = $\frac{N}{4}$ th placed item in the arranged data

Second Quartile (Q_2) = $\frac{N}{2}$ th placed item in the arranged data

Third Quartile (Q_3) = $3 \left(\frac{N}{4} \right)$ th placed item in the arranged data

For Grouped Data

If $x_1, x_2, \dots, x_k$ are k values (or mid values in case of class intervals) of a variable X with their corresponding frequencies $f_1, f_2, \dots, f_k$, then first of all we form a cumulative frequency distribution. After that we determine the i^{th} quartile class as similar as we do in case of median.

The i^{th} quartile is denoted by Q_i and defined as

$$Q_i = L + \frac{\left(\frac{iN}{4} - C \right)}{f} \times h \quad \text{for } i = 1, 2, 3$$

where, L = lower class limit of i^{th} quartile class,

h = width of the i^{th} quartile class,

N = total frequency,

C = cumulative frequency of pre i^{th} quartile class, and

f = frequencies of i^{th} quartile class.

“ i ” denotes i^{th} quartile class. It is the class in which $\left(\frac{i \times N}{4} \right)^{\text{th}}$ observation falls in cumulative frequency. It is easy to see that the second quartile ($i = 2$) is the median.

2.6.2 Deciles

Deciles divide whole distribution in to ten equal parts. There are nine deciles. $D_1, D_2, \dots, D_9$ are known as 1st Decile, 2nd Decile, ..., 9th Decile respectively and i^{th} Decile contains the $(iN/10)^{\text{th}}$ part of data. Here, it may be noted that the data should be arranged in ascending or descending order of magnitude.

For Ungrouped Data

For obtaining the deciles first of all we have to arrange the data either in ascending order or in descending order of their magnitude. Then, we find the $(1N/10)^{\text{th}}, (2N/10)^{\text{th}}, \dots, (9N/10)^{\text{th}}$ placed item in the arranged data for finding out the 1st decile (D_1), 2nd decile (D_2), ..., 9th decile (D_9) respectively. The value of the $(N/10)^{\text{th}}$ placed item would be the 1st decile, value of $(2N/10)^{\text{th}}$ placed item would be the 2nd decile. Similarly, the value of $(9N/10)^{\text{th}}$ placed item would be the 9th decile of that data.

Mathematically, if $x_1, x_2, \dots, x_N$ are N values of a variable X then the i^{th} decile is defined as:

$$i^{\text{th}} \text{ Decile } (D_i) = \frac{iN}{10} \text{th placed item in the arranged data } (i = 1, 2, 3, \dots, 9)$$

For Grouped Data

If $x_1, x_2, \dots, x_k$ are k values (or mid values in case of class intervals) of a variable X with their corresponding frequencies $f_1, f_2, \dots, f_k$, then first of all we form a cumulative frequency distribution. After that we determine the i^{th} deciles classes as similar as we do in case of quartiles.

The i^{th} decile is denoted by D_i and given by

$$D_i = L + \frac{\left(\frac{iN}{10} - C\right)}{f} \times h \quad \text{for } i = 1, 2, \dots, 9$$

where, L = lower class limit of i^{th} decile class,

h = width of the i^{th} decile class,

N = total frequency,

C = cumulative frequency of pre i^{th} decile class; and

f = frequency of i^{th} decile class.

“ i ” denotes i^{th} decile class. It is the class in which $\left(\frac{i \times N}{10}\right)^{\text{th}}$ observation falls in cumulative frequency. It is easy to see that the fifth quartile ($i=5$) is the median.

2.6.3 Percentiles

Percentiles divide whole distribution into 100 equal parts. There are ninety nine percentiles. $P_1, P_2, \dots, P_{99}$ are known as 1st percentile, 2nd percentile, ..., 99th percentile and i^{th} percentile contains the $(iN/100)^{\text{th}}$ part of data. Here, it may be noted that the data should be arranged in ascending or descending order of magnitude.

For Ungrouped Data

For obtaining the percentiles first of all we have to arrange the data either in ascending order or in descending order of their magnitude. Then, we find the $(1N/100)^{\text{th}}, (2N/100)^{\text{th}}, \dots, (99N/100)^{\text{th}}$ placed item in the arranged data for finding out the 1st percentile (P_1), 2nd percentile (P_2), ..., 99th percentile (P_{99}) respectively. The value of the $(N/100)^{\text{th}}$ placed item would be the 1st percentile, value of $(2N/100)^{\text{th}}$ placed item would be the 2nd percentile. Similarly, the value of $(99N/100)^{\text{th}}$ placed item would be the 99th percentile of that data.

Mathematically, if $x_1, x_2, \dots, x_N$ are N values of a variable X then the i^{th} percentile is defined as:

$$i^{\text{th}} \text{ Percentile } (P_i) = \frac{iN}{100} \text{th placed item in the arranged data } (i = 1, 2, \dots, 99)$$

For Grouped Data

If $x_1, x_2, \dots, x_k$ are k values (or mid values in case of class intervals) of a variable X with their corresponding frequencies $f_1, f_2, \dots, f_k$, then first of all we form a cumulative frequency distribution. After that we determine the i^{th} percentile class as similar as we do in case of median. The i^{th} percentile is denoted by P_i and given by

$$P_i = L + \frac{\left(\frac{iN}{100} - C\right)}{f} \times h \quad \text{for } i = 1, 2, \dots, 99$$

where, L = lower limit of i^{th} percentile class,

h = width of the i^{th} percentile class,

N = total frequency,

C = cumulative frequency of pre i^{th} percentile class; and

f = frequency of i^{th} percentile class.

“ i ” denotes i^{th} percentile class. It is the class in which $\left(\frac{i \times N}{100}\right)^{\text{th}}$ observation falls in cumulative frequency. It is easy to see that the fiftieth percentile ($i = 50$) is the median.

Example 16: For the data given in Example 8, calculate the first and third quartiles.

Solution: First we find cumulative frequency given in the following cumulative frequency table:

Class Interval	Frequency	Cumulative Frequency (< type)
0-10	3	3
10-20	5	8
20-30	7	15
30-40	9	24
40-50	4	28
	$N = \sum f = 28$	

Here, $N/4 = 28/4 = 7$. The 7th observation falls in the class 10-20. So, this is the first quartile class. $3N/4 = 21^{\text{th}}$ observation falls in class 30-40, so it is the third quartile class.

For first quartile $L = 10$, $f = 5$, $C = 3$, $N = 28$

$$Q_1 = 10 + \frac{(7-3)}{5} \times 10 = 18$$

For third quartile $L = 30$, $f = 9$, $C = 15$

$$Q_3 = 30 + \frac{(21-15)}{9} \times 10 = 36.67$$

2.6.4 Boxplots

In descriptive statistics, a box plot also known as a box-and-whisker plot, is a convenient way of graphically representing numerical data. It represents the data through five-number summary, i.e. the smallest observation (sample minimum), lower quartile (Q_1), median (Q_2), upper quartile (Q_3), and the largest observation (sample maximum). Box plots are used to describe a distribution generally when it is extremely skewed or multimodal. A box plot also indicates which observation(s), if any, might be considered as outliers. A box plot is a quick graphic approach for examining one or more sets of data.

Box plots display differences between populations without making any assumptions of the distributions. The spacing between the different parts of the box helps in indicating the degree of dispersion (spread) and skewness in the data, and identifies outliers. Box plots can be drawn either horizontally or vertically. Here we will draw box plots vertically.

Method of Construction of the Box Plots

Box plots are useful for identifying key values while comparing two or more distributions. To understand more clearly the method of constructing the box plots, let us consider the following data of 37 students in a class who were examined by a game with a box containing some balls. Their task was to select a ball from one box placed at one corner and put it in another blank box placed at another corner of the class as quickly as possible, and their times (in seconds) were recorded. The scores were compared for 16 boys and 21 girls who participated in a game. Observed data are given in the following table:

Time (in seconds) for completing the given task

Boys	18, 19, 20, 22, 24, 25, 26, 16, 17, 19, 25, 27, 28, 23, 23, 31
Girls	15, 17, 18, 19, 20, 21, 23, 14, 17, 18, 19, 20, 21, 24, 19, 16, 17, 18, 20, 22, 28

The method of construction of separate box plots for the data of boys and girls is discussed below:

There are several ways of constructing a box plot. The first relies on the quartiles, lowest and greatest values in the distribution of scores. Fig. 2.9 shows how these three statistics are used for the above example. We draw a box plot for each gender extending from the 1st quartile to the 3rd quartile. The 2nd quartile is drawn inside the box. Therefore,

- The bottom of each box is the 1st quartile,
- The top is the 3rd quartile,
- The line in the middle is the 2nd quartile.
- A line extending from the point corresponding to the smallest observation to 1st quartile is drawn and known as lower whisker.
- A line extending from the 3rd quartile to the point corresponding to the largest observation is drawn and is known as upper whisker.

Let us arrange the given data for each of the gender in ascending order as shown in following table to find out the above components, i.e. Q_1, Q_2, Q_3 , smallest observation and largest observation.

Gender	Times (in Seconds)
Boys	16, 17, 18, 19, 19, 20, 22, 22, 23, 23, 24, 25, 25, 27, 28, 31
Girls	14, 15, 16, 17, 17, 17, 18, 18, 18, 19, 19, 19, 20, 20, 20, 21, 21, 22, 23, 24, 28

For Boys

The lowest or smallest observation = $x_s = 16$.

$$\begin{aligned}\text{First quartile } (Q_1) &= \left(\frac{16+1}{4} \right)^{\text{th}} \text{ item} \\ &= 4.25^{\text{th}} \text{ item} = 4^{\text{th}} \text{ item} + 0.25 (5^{\text{th}} \text{ item} - 4^{\text{th}} \text{ item}) \\ &= 19 + 0.25 (19 - 19) = 19\end{aligned}$$

$$\begin{aligned}\text{Second quartile} = (Q_2) &= 2 \left(\frac{16+1}{4} \right)^{\text{th}} \text{ item} \\ &= 8.5^{\text{th}} \text{ item} = \text{Mean of } 8^{\text{th}} \text{ and } 9^{\text{th}} \text{ items} \\ &= \frac{22 + 23}{2} = 22.5\end{aligned}$$

$$\text{Third quartile } (Q_3) = 3 \left(\frac{16+1}{4} \right)^{\text{th}} \text{ item}$$

$$= 12.75^{\text{th}} \text{ item} = 12^{\text{th}} \text{ item} + 0.75 (13^{\text{th}} \text{ item} - 12^{\text{th}} \text{ item})$$

$$= 25 + 0.75 (25 - 25) = 25$$

The largest observation = $x_l = 31$

For Girls

The lowest or smallest observation = $x_s = 14$

$$\text{First quartile } (Q_1) = \left(\frac{21+1}{4} \right)^{\text{th}} \text{ item}$$

$$= 5.5^{\text{th}} \text{ item} = \text{mean of } 5^{\text{th}} \text{ and } 6^{\text{th}} \text{ items} = \frac{17+17}{2} = 17$$

$$\text{Second quartile } (Q_2) = 2 \left(\frac{21+1}{4} \right)^{\text{th}} \text{ item} = 11^{\text{th}} \text{ item} = 19$$

$$\text{Third quartile } (Q_3) = 3 \left(\frac{21+1}{4} \right)^{\text{th}} \text{ item}$$

$$= 16.5^{\text{th}} \text{ item} = \text{mean of } 16^{\text{th}} \text{ and } 17^{\text{th}} \text{ item} = \frac{21+21}{2} = 21$$

The largest observation = $x_l = 28$

Box plots for boys and girls on the basis of the above findings are shown below in Fig. 2.9.

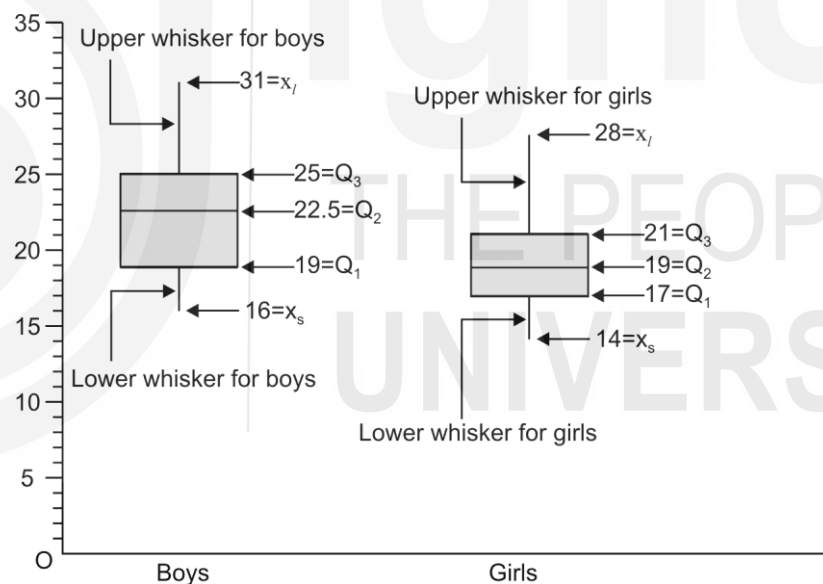


Fig. 2.9: The Box Plots with the Whiskers for Boys and Girls

Components of a Box Plot

Let us now discuss the various components and terminologies which are used in drawing the various types of box plots.

1. **Upper and Lower Hinges:** The upper and lower hinges are constructed to represent the third quartile and first quartile, respectively. For the example discussed above, the values of upper and lower hinges are 21 and 17 for the girls, whereas for the boys, they are 25 and 19, respectively.
2. **H-Spread:** This is calculated by taking the difference between the upper and lower hinges. The H-Spread shows the spread of the elements of the data between the first quartile and the third quartile. For the data related to boys

in the example discussed above, the H-spread is $25 - 19 = 6$, whereas for girls it is $21 - 17 = 4$.

3. **Whiskers:** The lines extending above and below the box are called whiskers. Lower whisker extends from the point corresponding to the smallest observation to Q_1 and the upper whisker extends from Q_3 to the largest observation.
4. **Step:** The step is calculated by multiplying the H-spread by 1.5. For the data of boys in the above example, the value of 1 step is $1.5 \times 6 = 9$, whereas for girls, it is $1.5 \times 4 = 6$.
5. **Upper and Lower Inner Fences:** The upper and lower inner fences are calculated by adding one step to the upper hinge and subtracting 1 step from the lower hinge, respectively. In other words, upper inner fence is equal to the 1 step ahead from upper hinge whereas the lower inner fence is one step before the lower hinge. For the data of boys in the example discussed above, the upper and lower inner fences are $25 + 9 = 34$ and $19 - 9 = 10$, whereas for the data of girls the upper and lower inner fences are $21 + 6 = 27$ and $17 - 6 = 11$, respectively.
6. **Upper and Lower Outer Fences:** The upper and lower outer fences are calculated by adding two steps to the upper hinge and subtracting 2 steps from the lower hinge, respectively. For the data of boys in the example discussed above the upper and lower outer fence are $25 + 2 \times 9 = 43$ and $19 - 2 \times 9 = 1$, whereas for the data of girls they are, $21 + 2 \times 6 = 33$ and $17 - 2 \times 6 = 5$, respectively.
7. **Upper and Lower Adjacent:** The upper and lower adjacent are used to represent the largest and the smallest observations of the data. For the data of boys in the example discussed above upper and lower adjacent are 31 and 16 whereas for the girls data they are 28 and 14, respectively.

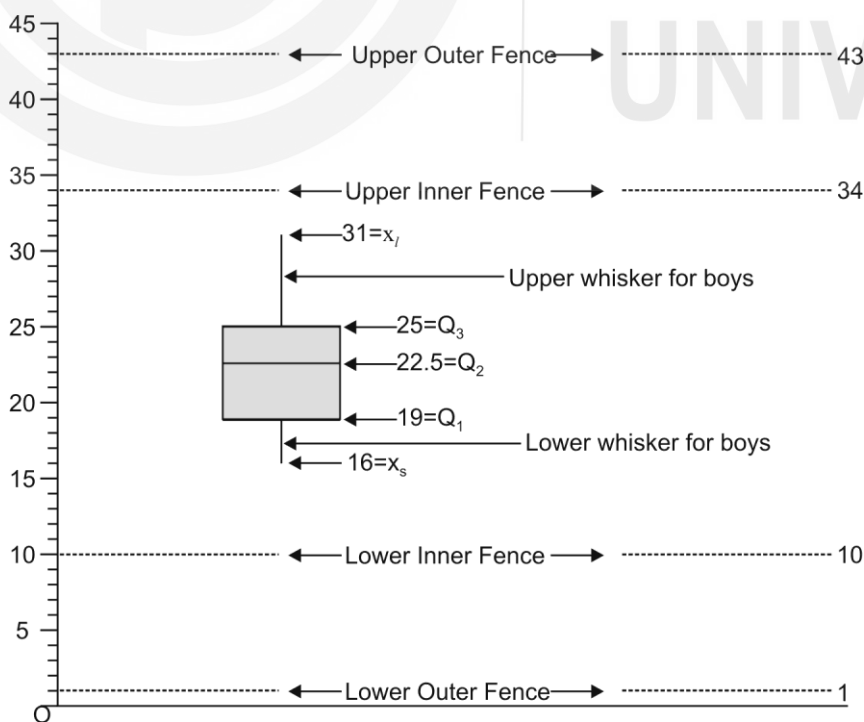


Fig. 2.10 Box Plot for Boys with Inner and Outer Fences

8. **Outside Value:** The outside value is a value which is beyond an inner fence but not beyond an outer fence. This is used to represent the scattered values of the data. To represent these values circles are used.
9. **Far Out Value:** A value which is beyond the upper outer fence or lower outer fence is known as far out value. To represent these values, the asterisks are used.

Box plot with the components discussed above for the data of boys is shown in Fig. 2.10.

Box plot with the components discussed above for the data of girls is shown in Fig. 2.11.

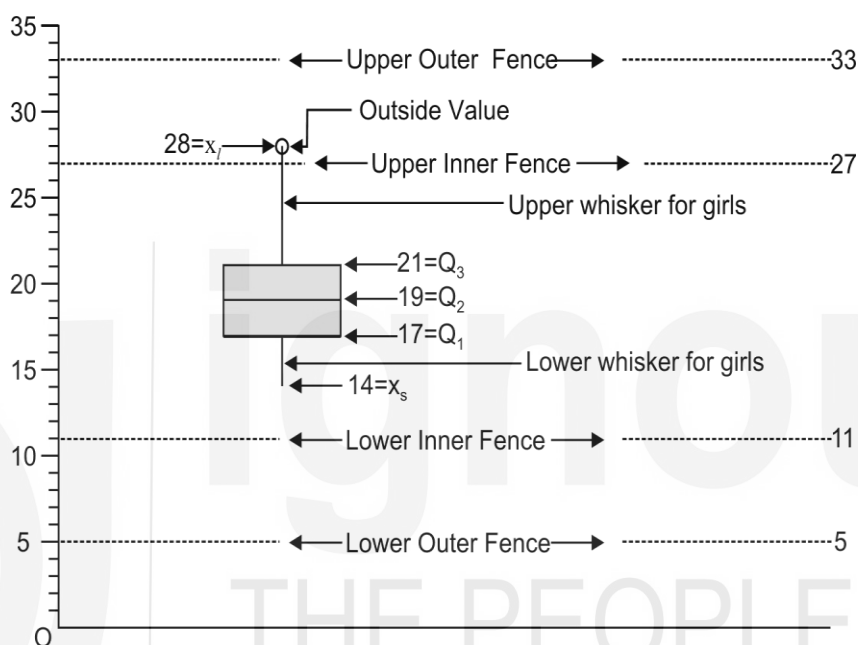


Fig. 2.11 Box Plot for Girls with Inner and Outer Fences

Box Plots with Outliers

Outside value(s) and far out value(s) are known as extreme observations. If the extreme observations are present in the data then those observations may be represented in box plot by an individual mark. The extreme observations are described as outliers when they are represented in box plots. The outside value is a value which is beyond an inner fence but not beyond an outer fence whereas the far out value is a value which is beyond the lower and upper outer fences. The individual marks for extreme values can be plotted above and below to the whiskers in box plot. Specially, outside values are indicated by small circles. The box plot for girls data indicating the outside value by a circle is shown in Fig. 2.12.

Box Plots with + Signs

One more component which is to be included in box plots are the value of some important parameters like, mean, mode, etc. We indicate the mean score for a group of values by inserting a plus sign in box plots. For the example mean in case of boys data is 20.875 and mean in case of girls data is 19.33, which are shown by a plus sign in Fig. 2.13.

Fig. 2.13 provides a revealing summary of the data. Since half of the scores are between the hinges (recall that the hinges are the first and third quartiles), we

see that half of the girl's times are between 17 and 21, whereas half of the boy's times are between 19 and 25.

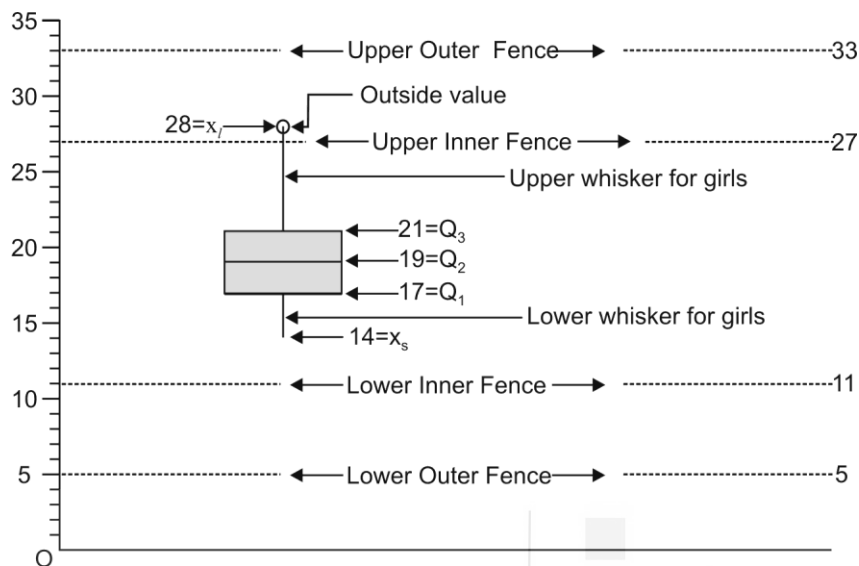


Fig. 2.12: The Box Blot for Girls with the Outlier.

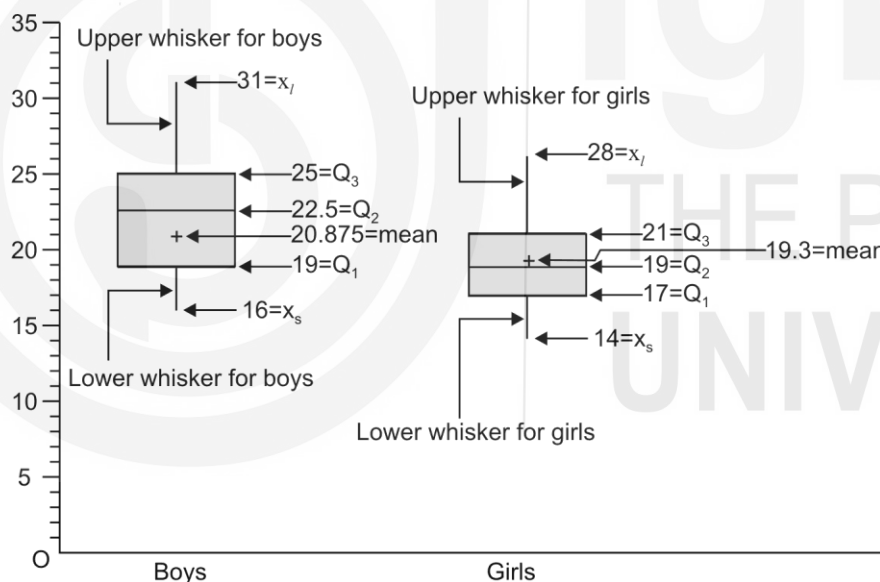


Fig. 2.13: The Box Plot with Whisker and + Signs for Boys and Girls Data.

Check Your Progress 4

- 1 Draw a box plot for the given data:
17, 15, 17, 20, 13, 15, 15, 16, 16, 15, 19, 12, 19, 14, 11, 14, 16, 10, 19, 18,
20, 14, 17, 19, 16, 22, 21, 23, 11, 12, 18, 13, 12, 25, 14, 15, 25, 17, 10, 21
- 2 Draw a box plot for the given data:
31, 42, 22, 27, 33, 27, 37, 28, 34, 44, 25, 39, 26, 31, 26, 33, 46, 48, 50

2.7 SUMMARY

In this unit, we have discussed:

1. How to make frequency distribution for different types of data.
 2. How to create class intervals for the given data.
 3. How to graphically represent frequency distribution of data.
 4. The properties of a good measures of central tendencies.
 5. How to compute different measures of central tendency.
 6. The different kinds of partition values and use of boxplots.
-

2.8 SOLUTIONS / ANSWERS

Check Your Progress 1

- 1 Let us determine the suitable class interval with the help of the following formula:

$$i = \frac{\text{Range}}{1 + 3.322 \log N}$$

$$\text{Range} = 59 - 06 = 53, N = 30$$

$$i = \frac{53}{1 + 3.322 \log 30} = \frac{53}{1 + 4.91} = 8.97 \approx 9$$

Since values like 3, 7, 9 etc., should be avoided and therefore, we will take 10 as the class interval and hence let us take the first class as 5-15 and thus the following table is formed:

Continuous Frequency Distribution of 30 Students

Heights (cm)	Tally Mark	Frequency
05-15		5
15-25		2
25-35		8
35-45		5
45-55		5
55-65		5
Total		30

- 2 As the least value is 3 and the highest value is 93, so using

$$i = \frac{\text{Range}}{1 + 3.322 \log N} = \frac{93-3}{1 + 3.322 \log 60} \approx 13.03 \approx 13$$

Since, values like 3, 7, 9, 11, 13 etc., should be avoided and therefore, we will take 14 as class interval and hence let us take the first class as 0-14 and thus the following table is formed.

Continuous Frequency Distribution of 60 Students According to their Heights

Profit (Cr)	Tally Mark	Frequency
0-14		06
15-29		04
30-44		16
45-59		10
60-74		14
75-89		07
90-104		03
Total		60

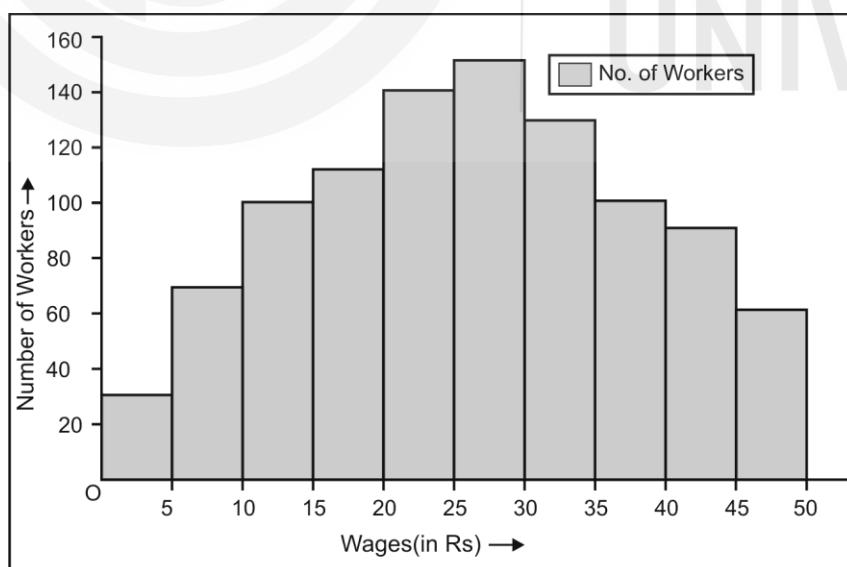
- 3 The following table illustrates the way of classification of data according to the exclusive method and principle of correction factor in classification.

Continuous Frequency Distribution of 60 Students According to their Heights

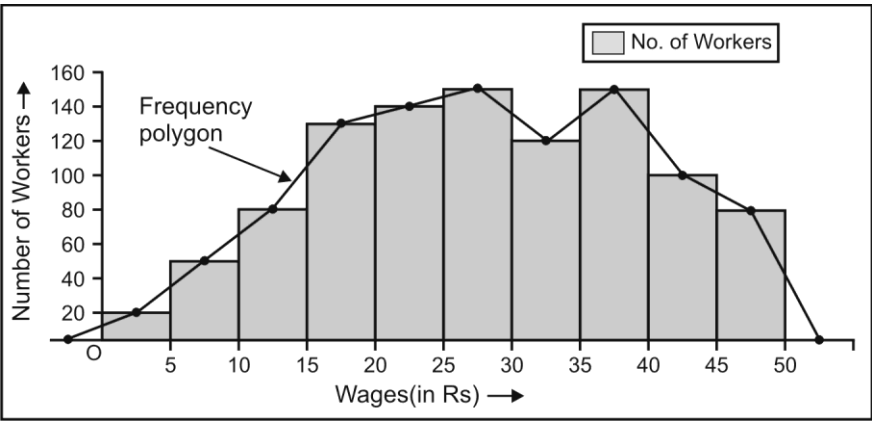
Profit (Cr)	Tally Mark	Frequency
0-14.5		06
14.5-29.5		04
29.5-44.5		16
44.5-59.5		10
59.5-74.5		14
74.5- 89.5		07
89.5-104.5		03
Total		60

Check Your Progress 2

- 1 Histogram of the given data is given below:



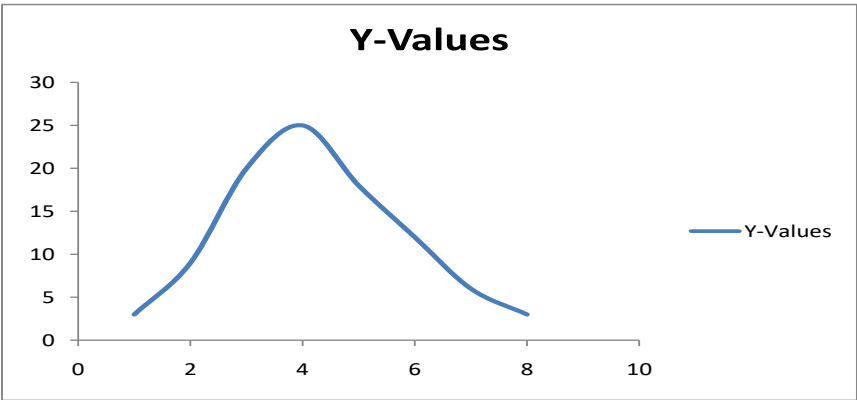
2 Frequency polygon of the given data by first drawing histogram is given below:



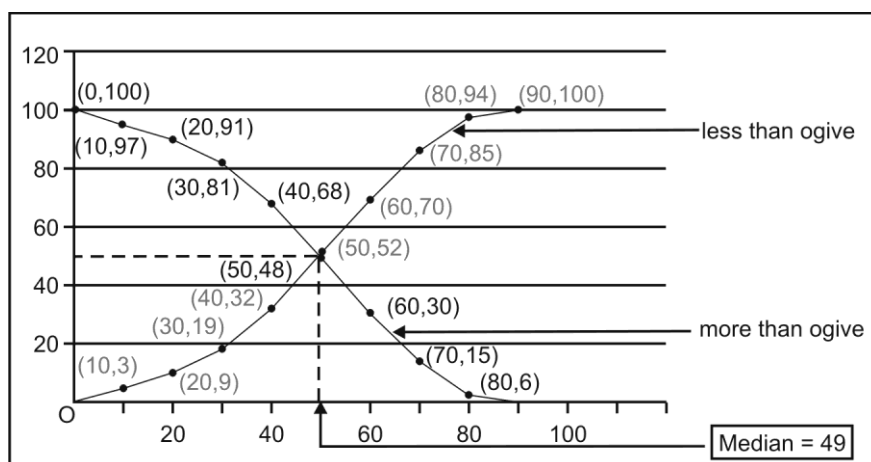
Frequency polygon can also be drawn without histogram as given below:



3 Frequency curve for the given data is given below.



4 Two ogives for the given data are given below:



5. Stem-and-leaf display of the given data of marks obtained by 30 students is given below.

```

5 | 9 7 0
6 | 8 5 1 2 1 9 4 7 2 5 5
7 | 7 0 0 3 6 9 9 1 7
8 | 0 2 7 0 2 3 0
    
```

After arranging the leaves in ascending order of magnitude, we have

```

5 | 0 7 9
6 | 1 1 2 2 4 5 5 5 7 8 9
7 | 0 0 1 3 6 7 7 9 9
8 | 0 0 0 2 2 3 7
    
```

Check Your Progress 3

1 We have the following frequency distribution:

Marks	No. of Students (f)	Mid Points x	fx
0-10	6	5	30
10-20	9	15	135
20-30	17	25	425
30-40	10	35	350
40-50	8	45	360
	$\sum f_i = 50$		$\sum f_i x_i = 1300$

$$\bar{X} = \frac{\sum_{i=1}^5 f_i x_i}{\sum_{i=1}^5 f_i} = \frac{1300}{50} = 26$$

Using short-cut method

Marks	f	x	$d = \frac{x - 25}{10}$	f d
0-10	6	5	-2	-12
10-20	9	15	-1	-9
20-30	17	25=A	0	0
30-40	10	35	+1	+10
40-50	8	45	+2	+16
	$\sum f = 50$			$\sum fd = 5$

Now
$$\bar{X} = A + \frac{\sum fd}{N} \times h$$

$$\begin{aligned}\bar{X} &= 25 + \frac{5}{50} \times 10 \\ &= 26\end{aligned}$$

2 First we shall calculate the cumulative frequency distribution

Class	f	Cumulative Frequency
0-20	5	5
20-40	15	20
40-60	30	50
60-80	f_4	$50 + f_4$
80-100	8	$58 + f_4$

Median class is 40-60 since median value is given 50,

$$\text{Median} = L + \frac{\left(\frac{N}{2} - C\right)}{f} \times h$$

$$50 = 40 + \frac{\left(\frac{58 + f_4}{2} - 20\right)}{30} \times 20$$

$$50 - 40 = \frac{58 + f_4 - 40}{3}$$

$$18 + f_4 = 30$$

$$f_4 = 30 - 18 = 12$$

3 We have

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

$$35.4 = 3 \text{ Median} - 2 (38.6)$$

$$3 \text{ Median} = 35.4 + 77.2$$

$$\text{Median} = 112.6/3 = 37.53$$

Check Your Progress 4

1 After arranging the given data in ascending order of magnitude, we have

10, 10, 11, 11, 12, 12, 12, 13, 13, 14, 14, 14, 14, 15, 15, 15, 15, 15, 16,
16, 16, 16, 17, 17, 17, 17, 18, 18, 19, 19, 19, 19, 20, 20, 21, 21, 22, 23,
25, 25

If you want, you can also present it in **descending order, table form**, or calculate **mean / median / frequency**.

The lowest or smallest observation = $x_s = 10$

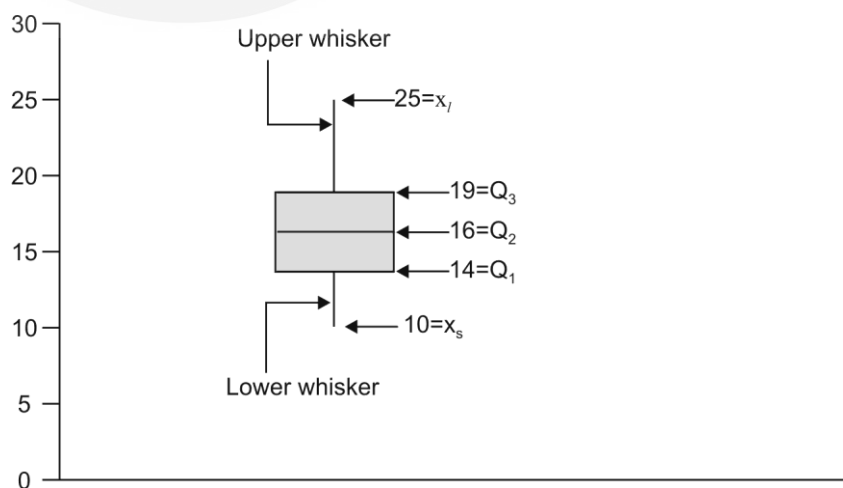
$$\begin{aligned}\text{First quartile } (Q_1) &= \left(\frac{40+1}{4} \right)^{\text{th}} \text{ item} \\ &= 10.25^{\text{th}} \text{ item} = 10^{\text{th}} \text{ item} + 0.25 (11^{\text{th}} \text{ item} - 10^{\text{th}} \text{ item}) \\ &= 14 + 0.25 (14 - 14) = 14\end{aligned}$$

$$\begin{aligned}\text{Second quartile } (Q_2) &= 2 \left(\frac{40+1}{4} \right)^{\text{th}} \text{ item} \\ &= 20.5^{\text{th}} \text{ item} = \text{Mean of } 20^{\text{th}} \text{ and } 21^{\text{st}} \text{ items} \\ &= \frac{16+16}{2} = 16\end{aligned}$$

$$\begin{aligned}\text{Third quartile } (Q_3) &= 3 \left(\frac{40+1}{4} \right)^{\text{th}} \text{ item} \\ &= 30.75^{\text{th}} \text{ item} = 30^{\text{th}} \text{ item} + 0.75 (31^{\text{st}} \text{ item} - 30^{\text{th}} \text{ item}) \\ &= 19 + 0.75 (19 - 19) = 19\end{aligned}$$

The largest observation = $x_l = 25$

Using above calculations box plot based on five-number summary (i.e. smallest observation x_s , first quartile (Q_1), second quartile (Q_2), third quartile Q_3), largest observation x_l) is given below:



2 After arranging the given data in ascending order of magnitude, we have

22, 25, 26, 26, 27, 27, 28, 31, 31, 33, 33, 34, 37, 39, 42, 44, 46, 48, 50

The lowest or smallest observation $= x_s = 22$

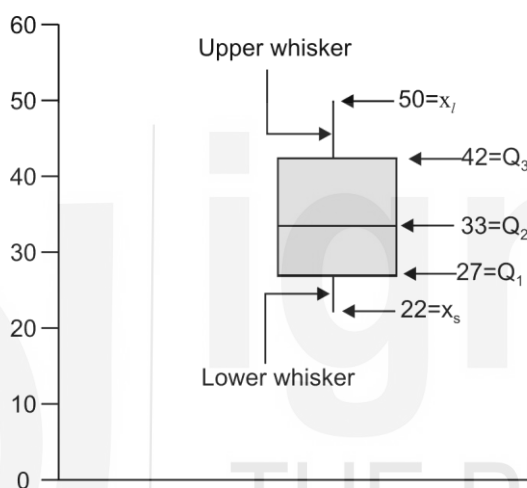
First quartile $(Q_1) = \left(\frac{19+1}{4}\right)^{\text{th}}$ item $= 5^{\text{th}}$ item $= 27$

Second quartile $(Q_2) = 2\left(\frac{19+1}{4}\right)^{\text{th}}$ item $= 10^{\text{th}}$ item $= 33$

Third quartile $(Q_3) = 3\left(\frac{19+1}{4}\right)^{\text{th}}$ item $= 15^{\text{th}}$ item $= 42$

The largest observation $= x_l = 50$

Using above calculations box plot based on five-number summary (i.e. smallest observation x_s , first quartile (Q_1) , second quartile (Q_2) , third quartile Q_3), largest observation x_l) is given below:



2.9 REFERENCES

IGNOU Course Material

1. Post-Graduate Diploma In Applied Statistics (PGDAST), MST 001: Matrices, Determinants and Collection of Data, UNIT 13: Classification of Data (for Sections 2.2, 2.3 and related portions in Sections 2.0, 2.1, 2.7, 2.8 and Check Your Progress questions)
2. PGDAST, MST 001: Matrices, Determinants and Collection of Data, UNIT 15: Graphical Representation of Data-I (for Section 2.4, 2.4.1 and related portions in Sections 2.0, 2.1, 2.7, 2.8 and Check Your Progress questions)
3. PGDAST, MST 001: Matrices, Determinants and Collection of Data, UNIT 16: Graphical Representation of Data -II (for Sections 2.4.2, 2.6.4 and related portions in Sections 2.0, 2.1, 2.7, 2.8 and Check Your Progress questions)
4. PGDAST, MST 002: Descriptive Statistics, UNIT 1: Measures of Central Tendency (for Sections 2.5, 2.6 to 2.6.3 and related portions in Sections 2.0, 2.1, 2.7, 2.8 and Check Your Progress questions)

Reference Book

5. "Introduction to Probability and Statistics for Engineers and Scientists", 3rd Edition, Sheldon M. Ross, Elsevier Academic Press, 2004

UNIT 3 MEASURES OF DISPERSION

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3.0 INTRODUCTION

Different measures of central tendency, discussed in Unit 2 of this block, give a value around which the data is concentrated. But it gives no idea about the nature of scatter or spread. For example, the observations 10, 30 and 50 have mean 30 while the observations 28, 30, 32 also have mean 30. Both the distributions are spread around 30. But it is observed that the variability among units is more in the first than in the second. In other words, there is greater variability or dispersion in the first set of observations in comparison to other. Measure of dispersion is calculated to get an idea about the variability in the data.

In Section 3.2, some basic measures of dispersion are described. Significance and properties of a good measure of dispersion are also explored. The basic idea about the range is explained in Section 3.2.1 whereas the measures of quartile deviation are introduced in Section 3.2.2. Mean deviation is described in Section 3.2.3. Methods of calculation of variance and standard deviation for grouped and ungrouped data are explained in Section 3.3 whereas in Section 3.3.3 the basis idea about root mean square deviation is provided. Coefficient of variation, which is a relative measure of variation is described in Section 3.3.4. Section 3.4 briefly defines the normal data distribution. Lack of symmetry of a frequency distribution is called skewness. If the distribution is not symmetric, the frequencies will not be normally distributed about the centre of the distribution. This unit also explains the concepts of skewness are described in Section 3.5 whereas the various measures of skewness are given with examples in Section 3.6. In Section 3.7, the concepts and a measure of kurtosis is discussed.

3.1 OBJECTIVES

After reading this unit, you would be able to

- conceptualise dispersion and its utility;
- explain the properties of a good measure of dispersion;
- explain methods of calculation of different types of measures of dispersion;
- solve the numerical problems related to the measures of dispersion.
- describe the properties of normal distribution;
- describe the concepts of skewness and its measures;
- explain the different measures of skewness;
- describe the concepts and a measure of kurtosis;
- explain how skewness and kurtosis describe the shape of a distribution.

3.2 BASIC MEASURES OF DISPERSION

According to Spiegel, the degree to which numerical data tend to spread about an average value is called the variation or dispersion of data. Actually, there are two basic kinds of measures of dispersion: (i) Absolute measures and (ii) Relative measures. The absolute measures of dispersion are used to measure the variability of a given data expressed in the same unit, while the relative measures are used to compare the variability of two or more sets of observations. Following are the different measures of dispersion:

1. Range
2. Quartile Deviation
3. Mean Deviation
4. Standard Deviation and Variance

What is the Significance of Measures of Dispersion?

Measures of dispersion are needed for the following four basic purposes:

1. Measures of dispersion determine the reliability of an average value. In other words, measures of variation are pointed out as to how far an average is representative of the entire data. When variation is less, the average closely represents the individual values of the data and when variation is large; the average may not closely represent all the units and be quite unreliable.
2. Another purpose of measuring variation is to determine the nature and causes of variations in order to control the variation itself. For example, the variation in the quality of product in the process form of production can be checked by quality control department by identifying the reason for the variations in the quality of product. Thus, measurements of dispersion are helpful to control the causes of variation.
3. The measures of dispersion enable us to compare two or more series with regard to their variability. The relative measures of dispersion may also determine the uniformity or consistency. Smaller value of relative measure of dispersion implies greater uniformity or consistency in the data.

- Measures of dispersion facilitate the use of other statistical methods. In other words, many powerful statistical tools in statistics such as correlation analysis, the testing of hypothesis, the analysis of variance, techniques of quality control, etc. are based on different measures of dispersion.

Properties of a Good Measure of Dispersion

A good measure of dispersion should possess the following properties:

- It should be simple to understand;
- It should be easy to compute;
- It should be rigidly defined;
- It should be based on each and every observation of data;
- It should be amenable to further algebraic treatment;
- It should have sampling stability; and
- It should not be unduly affected by extreme observations.

Different measures of dispersion are discussed in following sub-sections.

3.2.1 Range

Range is the simplest measure of dispersion. It is defined as the difference between the maximum value of the variable and the minimum value of the variable in the distribution. Its merit lies in its simplicity. The demerit is that it is a crude measure because it is using only the maximum and the minimum observations of variable. However, it still finds applications in Order Statistics and Statistical Quality Control. It can be defined as

$$R = X_{\text{Max}} - X_{\text{Min}}$$

where, X_{Max} : Maximum value of variable and

X_{Min} : Minimum value of variable.

Example 1: Find the range of the distribution 6, 8, 2, 10, 15, 5, 1, 13.

Solution: For the given distribution, the maximum value of variable is 15 and the minimum value of variable is 1. Hence range = 15 - 1 = 14.

Note: Range is the simplest measure of dispersion as it can be obtained from the largest and the smallest observations. It is used in statistical quality control. However, since it uses only the maximum and the minimum values of a variable in the series and gives no importance to other observations, it is not very suitable for algebraic treatment. Range is affected by fluctuations in sampling. For example, if a single value lower than the minimum or higher than the maximum is added or if the maximum or minimum value is deleted range is seriously affected.

Range is the measure having unit of the variable but is not a pure number. That's why sometimes coefficient of range is calculated by

$$\text{Coefficient of Range} = \frac{X_{\text{Max}} - X_{\text{Min}}}{X_{\text{Max}} + X_{\text{Min}}}$$

3.2.2 Quartile Deviation

You have already studied in about Q_1 and Q_3 , the first quartile and the third quartile respectively in the Unit 2. ($Q_3 - Q_1$) gives the interquartile range. The

semi-interquartile range which is also known as Quartile Deviation (QD) is given by

$$\text{Quartile Deviation (QD)} = (Q_3 - Q_1) / 2$$

Relative measure of Q.D. known as Coefficient of Q.D. and is defined as

$$\text{Coefficient of QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Example 2: For the following data, find the quartile deviation:

Class Interval	0-10	10-20	20-30	30-40	40-50
Frequency	03	05	07	09	04

Solution: We have $N/4 = 28/4 = 7$ and 7th observation falls in the class 10-20. This is the first quartile class.

Similarly, $3N/4 = 21$ and 21st observation falls in the interval 30-40. This is the third quartile class.

Class Interval	Frequency	Cumulative Frequency
0-10	3	3
10-20	5	8
20-30	7	15
30-40	9	24
40-50	4	28

Using the formulae of first quartile and third quartile we found

$$Q_1 = 10 + \frac{(7-3)}{5} \times 10 = 18$$

$$Q_3 = 30 + \frac{(21-15)}{9} \times 10 = 36.67$$

Hence

$$\text{Quartile Deviation} = (36.67 - 18) / 2 = 9.335$$

3.2.3 Mean Deviation

Mean deviation is defined as average of the sum of the absolute values of deviation from any arbitrary value viz. mean, median, mode, etc. It is often suggested to calculate it from the median because it gives least value when measured from the median.

The deviation of an observation x_i from the assumed mean A is defined as $(x_i - A)$.

Therefore, the mean deviation can be defined as

$$MD = \frac{1}{n} \sum_{i=1}^n |x_i - A|$$

The quantity $\sum |x_i - A|$ is minimum when A is median.

We accordingly define mean deviation from mean as

$$MD = \frac{\sum_{i=1}^n |x_i - \bar{x}|}{n}$$

and from the median as

$$MD = \frac{\sum_{i=1}^n |x_i - \text{median}|}{n}$$

For frequency distribution, the formula will be

$$MD = \frac{\sum_{i=1}^k f_i |x_i - \bar{x}|}{\sum_{i=1}^k f_i}$$

$$MD = \frac{\sum_{i=1}^k f_i |x_i - \text{median}|}{\sum_{i=1}^k f_i}$$

where, all symbols have usual meanings.

Example 3: Find mean deviation for the given data

1, 2, 3, 4, 5, 6, 7

Solution: First of all we find Mean

$$\bar{x} = \frac{1+2+3+4+5+6+7}{7} = \frac{28}{7} = 4$$

Then, we will find $|x_i - \bar{x}|$: 3, 2, 1, 0, 1, 2, 3

So, $\sum |x_i - \bar{x}| = 12$

Therefore,

$$MD = \frac{12}{7} = 1.71$$

Example 4: Find mean deviation from mean for the following data:

x	1	2	3	4	5	6	7
f	3	5	8	12	10	7	5

Solution: First of all we have to calculate the mean from the given data

x	f	fx	$ x - \bar{x} $	$f x - \bar{x} $
1	3	3	3.24	9.72
2	5	10	2.24	11.20
3	8	24	1.24	9.92
4	12	48	0.24	2.88
5	10	50	0.76	7.60
6	7	42	1.76	12.32
7	5	35	2.76	13.80
	50	212	12.24	67.44

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{212}{50} = 4.24$$

$$MD = \frac{\sum_{i=1}^k f_i |x_i - \bar{x}|}{\sum_{i=1}^k f_i} = \frac{67.44}{50} = 1.348$$

Example 5: Consider the data given in Example 2 and find the Mean deviation from median.

Solution:

Class	Mid Value (x)	Frequency (f)	C.F.	$f_i x_i - \text{median} $
0-10	5	3	3	70.71
10-20	15	5	8	67.85
20-30	25	7	15	24.99
30-40	35	9	24	57.87
40-50	45	4	28	65.72
		$\Sigma f = 28$		$\Sigma f_i x_i - \text{median} = 287.14$

We have $N/2 = 28/2 = 14$

The 14th observation falls in the class 20-30. This is therefore the median class.

Using the formula of median, $\text{Median} = 20 + \frac{14-8}{7} \times 10 = 28.57$

$$\text{Mean Deviation from Median} = \frac{\sum_{i=1}^k f_i |x_i - \text{median}|}{\sum_{i=1}^k f_i} = 10.255$$

Note: Mean deviation utilises all the observations and is easy to understand and calculate. It is not much affected by extreme values. However, while calculating mean deviation, the negative deviations are made positive, thus, mean deviation is not amenable to algebraic treatment.

Check Your Progress 1:

1) Calculate the range for the following data:

60, 65, 70, 12, 52, 40, 48

.....

.....

2) Calculate range for the following frequency distribution:

Class	0-10	10-20	20-30	30-40	40-50
Frequency	2	6	12	7	3

.....

.....

3) Calculate the Quartile Deviation for the following data:

Class	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40
Frequency	6	8	12	24	36	32	24	8

.....

.....

.....

.....

4) Following are the marks of 7 students in Statistics:

16, 24, 13, 18, 15, 10, 23

Find the mean deviation from mean.

.....

.....

.....

.....

5) Find mean deviation for the following distribution:

Class	0-10	10-20	20-30	30-40	40-50
Frequency	5	8	15	16	6

.....

.....

.....

.....

3.3 VARIANCE AND STANDARD DEVIATION

In this section, we discuss one of the most used measures of dispersion and certain related measures. We will first discuss the variance, followed by the standard deviation and the root mean square variation.

3.3.1 Variance

Variance is the average of the square of deviations of the values taken from mean. Taking a square of the deviation is a better technique to get rid of negative deviations.

Variance is defined as

$$\text{Var}(x) = \sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

and for a frequency distribution, the formula is

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^k f_i (x_i - \bar{x})^2$$

where, all symbols have their usual meanings.

It should be noted that sum of squares of deviations is least when deviations are measured from the mean. This means $\sum (x_i - A)^2$ is least when $A = \text{Mean}$.

Example 6: Calculate the variance for the data given in Example 3.

Solution:

x	$(x - \bar{x})$	$(x - \bar{x})^2$
1	-3	9
2	-2	4
3	-1	1
4	0	0
5	1	1
6	2	4
7	3	9

We have $\sum_{i=1}^n (x_i - \bar{x})^2 = 28$

Therefore, $\text{Var}(x) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{7} \times 28 = 4$

Example 7: For the data given in Example 2, compute the variance.

Solution: We have the following data:

Class	Mid Value	Frequency (f)	$f(x - \bar{x})^2$
0-10	5	3	1470.924
10-20	15	5	737.250
20-30	25	7	32.1441
30-40	35	9	555.606
40-50	45	4	1275.504
		$\sum f = 28$	$\sum f(x_i - \bar{x})^2$ =4071.428

$$\text{Variance} = \frac{\sum_{i=1}^k f_i (x_i - \bar{x})^2}{\sum_{i=1}^k f_i}$$

$$= 4071.429 / 28 = 145.408$$

Note: A Variance is a rigidly defined measure that utilises all the observations. It is amenable to algebraic treatment and get rid of negative deviations by squaring. It is the most popular measure of dispersion. However, in cases where mean is not a suitable average, like when open ended classes are present, variance may not be the coveted measure of dispersion. In such cases quartile deviation may be used. Further, its unit is the square of the unit of the variable, due to which it is difficult to judge the magnitude of dispersion compared to standard deviation.

Variance of Combined Series

If there are two or more populations and the information about the means and variances of those populations are available then we can obtain the combined variance of several populations. If $n_1, n_2, \dots, n_k$ are the sizes, $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_k$ are the means and $\sigma_1^2, \sigma_2^2, \dots, \sigma_k^2$ are the variances of k populations, then the combined variance is given by

$$\sigma^2 = \frac{1}{(n_1 + n_2 + \dots + n_k)} [n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2) + \dots + n_k(\sigma_k^2 + d_k^2)]$$

where, $d_i = \bar{x}_i - \bar{x}$ and

$$\bar{x} = \frac{\sum_{i=1}^k n_i \bar{x}_i}{\sum_{i=1}^k n_i} \text{ is the mean of the combined series.}$$

Example 8: Suppose a series of 100 observations has mean 50 and variance 20. Another series of 200 observations has mean 80 and variance 40. What is the combined variance of the given series?

Solution: First we find the mean of the combined series

$$\begin{aligned} \bar{x} &= \frac{(100 \times 50 + 200 \times 80)}{(100 + 200)} \\ &= \frac{21000}{300} = 70 \end{aligned}$$

Therefore, $d_1 = 50 - 70 = -20$ and $d_2 = 80 - 70 = 10$

Variance of the combined series

$$\begin{aligned} &= \frac{(100 \times (20 + 400) + 200 \times (40 + 100))}{(100 + 200)} \\ &\Rightarrow \frac{(42000 + 28000)}{(300)} = \frac{70000}{300} = 233.33 \end{aligned}$$

3.3.2 Standard Deviation

Standard deviation (SD) is defined as the positive square root of variance. The formula is

$$SD = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$$

and for a frequency distribution the formula is

$$SD = \sqrt{\frac{\sum_{i=1}^k f_i (x_i - \bar{x})^2}{\sum_{i=1}^k f_i}}$$

where, all symbols have usual meanings. SD, MD and variance cannot be negative.

Example 9: Find the SD for the data in Example 2.

Solution: In the Example 7, we have already found the

$$\text{Variance} = 145.408$$

So

$$\text{SD} = +\sqrt{(145.408)} = 12.04$$

Note: Standard deviation is a rigidly defined measure that utilises all the observations. It is amenable to algebraic treatment and get rid of negative deviations by squaring. It is the most popular measure of dispersion. However, in cases where mean is not a suitable average, like when open ended classes are present, variance may not be the coveted measure of dispersion. In such cases quartile deviation may be used.

Remark:

SD is algebraically more amenable than MD. MD straightaway neglects the sign of observations which is also a useful information. In SD, the signs are considered but they get automatically eliminated due to squaring. However, MD will be useful in situations where Median is more suitable than mean as an average.

3.3.3 Root Mean Square Deviation

As we have discussed in the last sub-section, standard deviation is the positive square root of the average of the squares of deviations taken from the mean. If we take the deviations from assumed mean then it is called Root Mean Square Deviation and it is defined as

$$\text{RMSD} = \sqrt{\frac{\sum_{i=1}^n (x_i - A)^2}{n}}$$

where, A is the assumed mean.

For a frequency distribution, the formula is

$$\text{RMSD} = \sqrt{\frac{\sum_{i=1}^k f_i (x_i - A)^2}{\sum_{i=1}^k f_i}}$$

When assumed mean is equal to the actual mean $\bar{x}$ i.e. $A = \bar{x}$ root mean square deviation will be equal to the standard deviation.

3.3.4 Coefficient Of Variation

Coefficient of Variation (CV) is defined as

$$\text{CV} = \frac{\sigma}{\bar{x}} \times 100$$

It is a relative measure of variability. If we are comparing the two data series, the data series having smaller CV will be more consistent. One should be careful in making interpretation with CV. For example, the series 10, 10, 10 has SD zero and hence CV is also zero. The series 50, 50 and 50 also has SD zero and hence

CV is zero. But the second series has higher mean. So we shall regard the second series as more consistent than the first.

Example 10: Suppose batsman A has mean 50 with SD 10. Batsman B has mean 30 with SD 3. What do you infer about their performance?

Solution: A has higher mean than B. This means A is a better run maker.

However, B has lower CV ($3/30 = 0.1$) than A ($10/50 = 0.2$) and is consequently more consistent.

Check Your Progress 2

- 1) For a group containing 100 observations the arithmetic mean and standard deviation are 16 and $\sqrt{21}$ respectively. For 50 observations, the mean and standard deviation are 20 and 2 respectively. Calculate mean and standard deviation of other half.

.....

- 2) Find the standard deviation for the following numbers:

10 27 40 60 33 30 10

.....

- 3) Calculate standard deviation for the following data:

Class	0-10	10-20	20-30	30-40	40-50
Frequency	5	8	15	16	6

.....

- 4) If $n = 10$, $\bar{x} = 4$, $\sum x^2 = 200$, find the coefficient of variation.

.....

- 5) For a distribution, the coefficient of variation is 70.6% and the value of arithmetic mean is 16. Find the value of standard deviation.

.....

3.4 NORMAL DATA SETS

The frequency distribution of the normal data is of the shape of a 'Bell' and is symmetrical on both sides. This bell-shaped curve, also known as the Normal Curve, has identical values of all three measures of central tendency (Mean, Median and Mode). This frequency distribution is known as the Normal Distribution. Many data variables, like height, weight, age, blood sugar level, etc., are found to be normally distributed in nature.

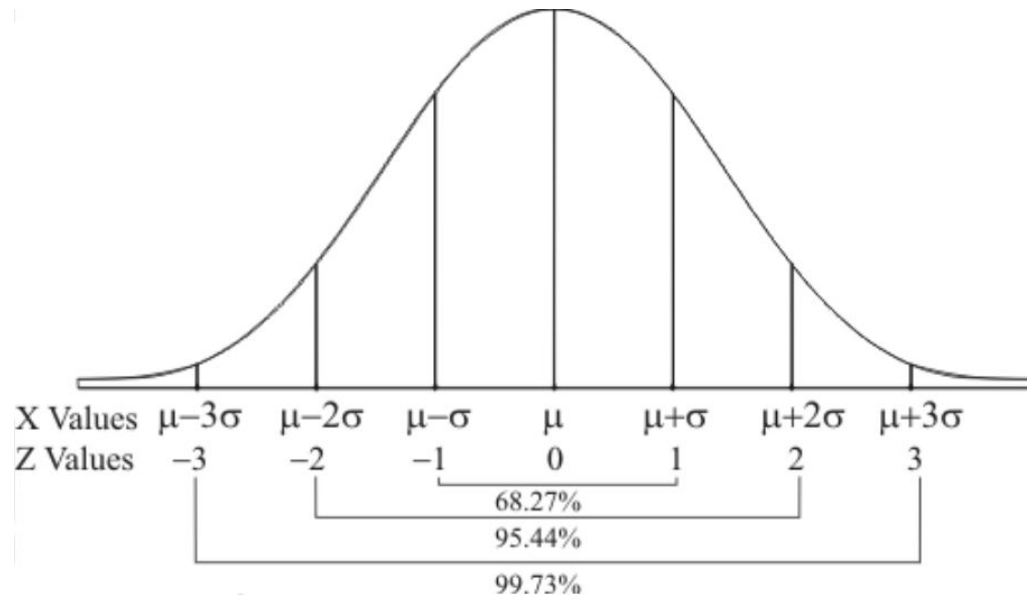


Fig. 3.1: Normal Distribution

The following are the characteristics of a normal distribution:

1. The Normal Curve is Symmetrical: The normal probability curve is symmetrical around its vertical axis, called the ordinate. The symmetry about the ordinate at the central point of the curve implies that the left and right halves of the middle central point are mirror images (Refer to Fig. 3.1).
2. The Normal Curve is Unimodal: Since there is only one maximum point in the curve, the normal probability curve is unimodal, i.e. it has only one mode.
3. The Maximum Ordinate occurs at the Centre: The maximum height of the ordinate always occurs at the central point of the curve, that is, the mid-point.
4. The Normal Curve is Asymptotic to the X Axis: The normal probability curve approaches the horizontal axis asymptotically; i.e. the curve continues to decrease in height on both ends away from the middle point (the maximum ordinate point), but it never touches the horizontal axis. Therefore, its ends extend from minus infinity ($-\infty$) to plus infinity ($+\infty$).
5. The Height of the Curve declines Symmetrically: In the normal probability curve, the height declines symmetrically in either direction from the maximum point.
6. The Points of Influx occur at point ± 1 Standard Deviation ($\pm 1 \sigma$): The normal curve changes its direction from convex to concave at a point recognised as point of influx.
7. The Total Percentage of Area of the Normal Curve within Two Points of inflexion is fixed: Approximately 68.26 of % area of the curve lies within the limits of ± 1 standard deviation ($\pm 1 \sigma$) unit from the mean.
8. The Total Area under Normal Curve may also be considered 100 Per cent Probability: The total area under the normal curve may be considered to approach 100 per cent probability, interpreted in terms of standard deviations.
9. The Normal Curve is Bilateral: The 50% area of the curve lies to the left side of the maximum central ordinate, and 50% of the area lies to the right side. Hence, the curve is bilateral.

Any deviation from the normal distribution of data is measured by the skewness and kurtosis, which are defined in the subsequent sections.

3.5 CONCEPT OF SKEWNESS

Skewness means lack of symmetry. In mathematics, a figure is called symmetric if there exists a point in it through which if a perpendicular is drawn on the X-axis, it divides the figure into two congruent parts i.e. identical in all respect or one part can be superimposed on the other i.e. mirror images of each other. In Statistics, a distribution is called symmetric if mean, median and mode coincide. Otherwise, the distribution becomes asymmetric. If the right tail is longer, we get a positively skewed distribution for which $\text{mean} > \text{median} > \text{mode}$ while if the left tail is longer, we get a negatively skewed distribution for which $\text{mean} < \text{median} < \text{mode}$.

The example of the Symmetrical curve, Positive skewed curve and Negative skewed curve are given as follows:

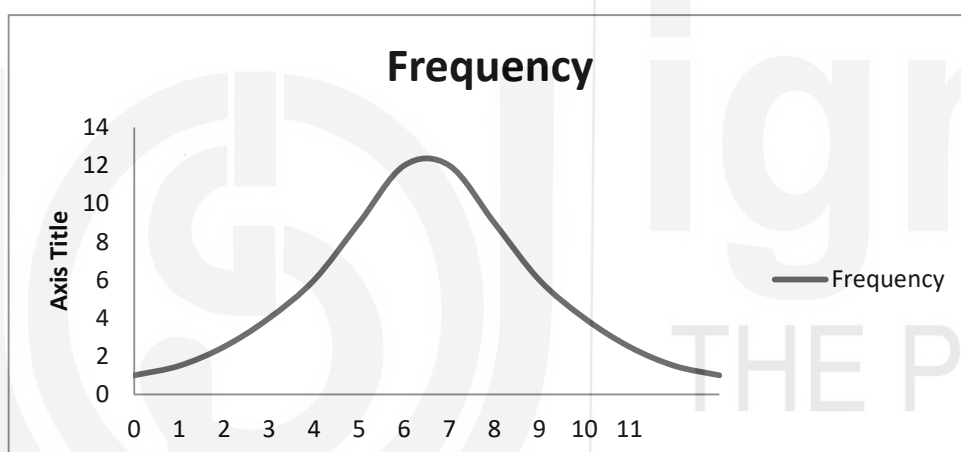


Fig. 3.2: Symmetrical Curve

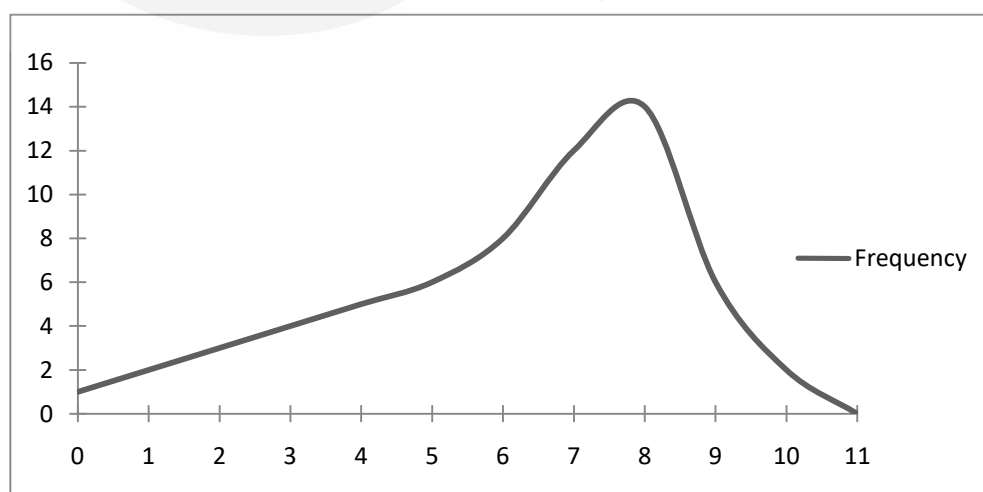


Fig. 3.3: Negative Skewed Curve

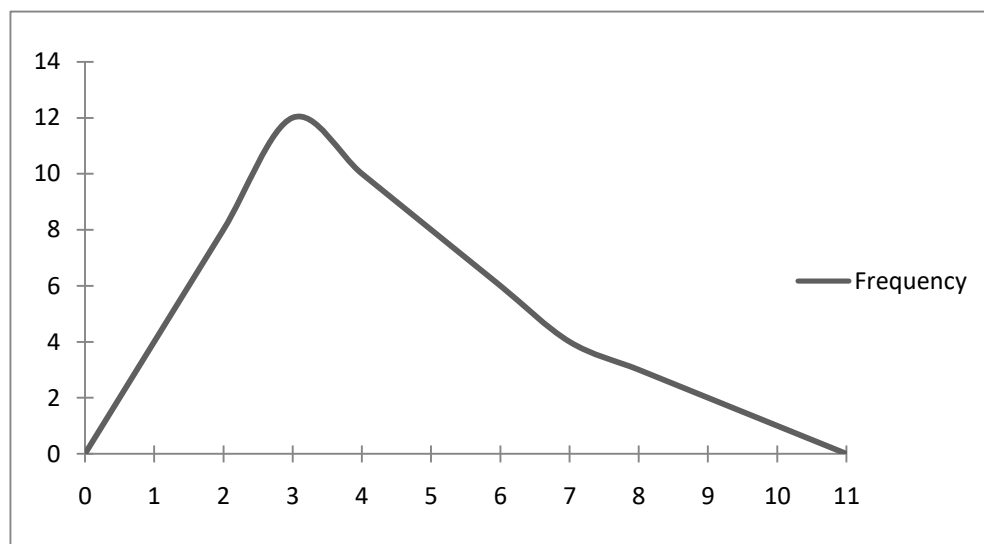


Fig. 3.4: Positive Skewed Curve

Difference between Variance and Skewness

The following two points of difference between variance and skewness should be carefully noted.

1. Variance tells us about the amount of variability while skewness gives the direction of variability.
2. In business and economic series, measures of variation have greater practical application than measures of skewness. However, in medical and life science field measures of skewness have greater practical applications than the variance.

3.6 VARIOUS MEASURES OF SKEWNESS

Measures of skewness help us to know to what degree and in which direction (positive or negative) the frequency distribution has a departure from symmetry. Although positive or negative skewness can be detected graphically depending on whether the right tail or the left tail is longer but, we don't get idea of the magnitude. Besides, borderline cases between symmetry and asymmetry may be difficult to detect graphically. Hence some statistical measures are required to find the magnitude of lack of symmetry. A good measure of skewness should possess three criteria:

1. It should be a unit free number so that the shapes of different distributions, so far as symmetry is concerned, can be compared even if the unit of the underlying variables are different;
2. If the distribution is symmetric, the value of the measure should be zero. Similarly, the measure should give positive or negative values according as the distribution has positive or negative skewness respectively; and
3. As we move from extreme negative skewness to extreme positive skewness, the value of the measure should vary accordingly.

Measures of skewness can be both absolute as well as relative. Since in a symmetrical distribution mean, median and mode are identical more the mean moves away from the mode, the larger the asymmetry or skewness. An absolute measure of skewness cannot be used for purposes of comparison because of the

same amount of skewness has different meanings in distribution with small variation and in distribution with large variation.

Absolute Measures of Skewness

Following are the absolute measures of skewness:

1. Skewness (S_k) = Mean – Median
2. Skewness (S_k) = Mean – Mode
3. Skewness (S_k) = ($Q_3 - Q_2$) - ($Q_2 - Q_1$)

For comparing to series, we do not calculate these absolute measures we calculate the relative measures which are called coefficient of skewness. Coefficient of skewness are pure numbers independent of units of measurements.

Relative Measures of Skewness

In order to make valid comparison between the skewness of two or more distributions we have to eliminate the distributing influence of variation. Such elimination can be done by dividing the absolute skewness by standard deviation. Skewness can be measured using the β and γ coefficients of Skewness, which were defined by Karl Pearson based upon the second and third central moments. Discussion on these is beyond the scope of this Unit and can be studied from the references.

In this section we discuss few simpler ways of defining Skewness.

1. Karl Pearson's Coefficient of Skewness

This method is most frequently used for measuring skewness. The formula for measuring coefficient of skewness is given by

$$S_k = \frac{\text{Mean} - \text{Mode}}{\sigma}$$

The value of this coefficient would be zero in a symmetrical distribution. If mean is greater than mode, coefficient of skewness would be positive otherwise negative. The value of the Karl Pearson's coefficient of skewness usually lies between ± 1 for moderately skewed distribution. If mode is not well defined, we use the formula

$$S_k = \frac{3(\text{Mean} - \text{Median})}{\sigma}$$

By using the relationship

$$\text{Mode} = (3 \text{ Median} - 2 \text{ Mean})$$

Here, $-3 \leq S_k \leq 3$. In practice it is rarely obtained.

2. Bowleys's Coefficient of Skewness

This method is based on quartiles. The formula for calculating coefficient of skewness is given by

$$S_k = \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{(Q_3 - Q_1)}$$

$$= \frac{(Q_3 - 2Q_2 + Q_1)}{(Q_3 - Q_1)}$$

The value of S_k would be zero if it is a symmetrical distribution. If the value is greater than zero, it is positively skewed and if the value is less than zero it is negatively skewed distribution. It will take value between +1 and -1.

3. Kelly's Coefficient of Skewness

The coefficient of skewness proposed by Kelly is based on percentiles and deciles. The formula for calculating the coefficient of skewness is given by

Based on Percentiles

$$S_k = \frac{(P_{90} - P_{50}) - (P_{50} - P_{10})}{(P_{90} - P_{10})}$$

$$= \frac{(P_{90} - 2P_{50} + P_{10})}{(P_{90} - P_{10})}$$

where, P_{90} , P_{50} and P_{10} are 90th, 50th and 10th Percentiles.

Based on Deciles

$$S_k = \frac{(D_9 - 2D_5 + D_1)}{D_9 - D_1}$$

where, D_9 , D_5 and D_1 are 9th, 5th and 1st Decile.

Example 11: For a distribution, Karl Pearson's coefficient of skewness is 0.64, standard deviation is 13, and mean is 59.2. Find the mode and median.

Solution: We have given

$$S_k = 0.64, \sigma = 13 \text{ and Mean} = 59.2$$

Therefore by using formulae

$$S_k = \frac{\text{Mean} - \text{Mode}}{\sigma}$$

$$0.64 = \frac{59.2 - \text{Mode}}{13}$$

$$\text{Mode} = 59.20 - 8.32 = 50.88$$

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

$$50.88 = 3 \text{ Median} - 2 (59.2)$$

$$\text{Median} = \frac{50.88 + 118.4}{3} = \frac{169.28}{3} = 56.42$$

Remarks about Skewness

1. If the value of mean, median and mode are same in any distribution, then the skewness does not exist in that distribution. Larger the difference in these values, larger the skewness;
2. If sum of the frequencies are equal on the both sides of mode then skewness does not exist;
3. If the distance of first quartile and third quartile are same from the median then a skewness does not exist. Similarly if deciles (first and ninth) and

percentiles (first and ninety nine) are at equal distance from the median. Then there is no asymmetry;

4. If the sums of positive and negative deviations obtained from mean, median or mode are equal then there is no asymmetry; and
5. If a graph of a data become a normal curve and when it is folded at middle and one part overlap fully on the other one then there is no asymmetry.

3.7 CONCEPT OF KURTOSIS

If we have the knowledge of the measures of central tendency, dispersion and skewness, even then we cannot get a complete idea of a distribution. In addition to these measures, we need to know another measure to get the complete idea about the shape of the distribution which can be studied with the help of Kurtosis. Prof. Karl Pearson has called it the “Convexity of a Curve”. Kurtosis gives a measure of flatness of distribution.

The degree of kurtosis of a distribution is measured relative to that of a normal curve. The curves with greater peakedness than the normal curve are called “**Leptokurtic**”. The curves which are more flat than the normal curve are called “**Platykurtic**”. The normal curve is called “**Mesokurtic**.” The Fig. 3.5 describes the three different curves mentioned above:

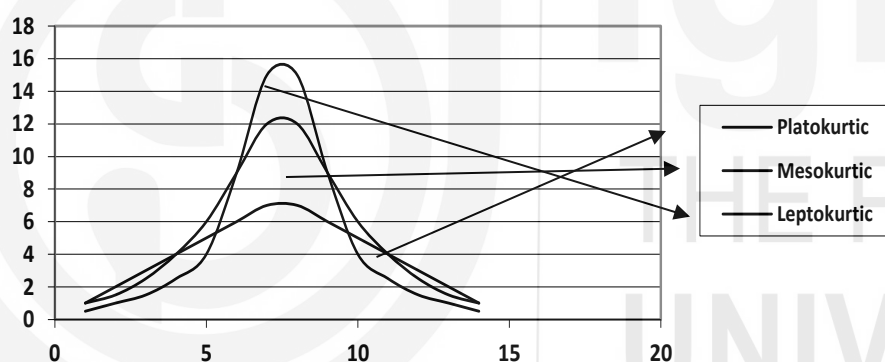


Fig. 3.5: Platykurtic Curve, Mesokurtic Curve and Leptokurtic Curve

Here we discuss one simple measure of Kurtosis. You can refer to other measures of kurtosis from references.

Kelly's Measure of Kurtosis

Kelly has given a measure of kurtosis based on percentiles. The formula is given by

$$\beta_2 = \frac{P_{75} - P_{25}}{P_{90} - P_{10}}$$

where, P_{75} , P_{25} , P_{90} , and P_{10} are 75th, 25th, 90th and 10th percentiles of dispersion respectively.

If $\beta_2 > 0.26315$, then the distribution is platykurtic.

If $\beta_2 < 0.26315$, then the distribution is leptokurtic.

Check Your Progress 3

- 1) Karl Pearson's coefficient of skewness is 1.28, its mean is 164 and mode 100, find the standard deviation.

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- 2) For a frequency distribution the Bowley's coefficient of skewness is 1.2. If the sum of the 1st and 3rd quarterlies is 200 and median is 76, find the value of third quartile.

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- 3) The following are the marks of 150 students in an examination. Calculate Karl Pearson's coefficient of skewness.

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No. of Students	10	40	20	0	10	40	16	14

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3.8 SUMMARY

In this unit, we have discussed:

1. A point to remember - While the mean value gives the value around which a distribution is scattered, the measure of dispersion tells how it is scattered.
2. This unit discusses the measures of dispersion, its use and measures of dispersion
3. The properties of a good measure of dispersion are also explained.
4. The different types of measures of dispersion were also discussed.
5. A suitable measure of central tendency and dispersion is calculated (say, arithmetic mean and SD), we get a good idea of the distribution even if the distribution is large.
6. This unit also defines the normal data and properties of normal distribution
7. It defines the significance of skewness and different kinds of measures of skewness.
8. The unit also explains the concept of kurtosis and different measures of kurtosis

3.9 SOLUTIONS /ANSWERS**Check Your Progress 1**

- 1) Maximum value (X_{Max}) = 70
 Minimum value (X_{Min}) = 12
 Therefore, $R = X_{\text{Max}} - X_{\text{Min}}$
 $= 70 - 12 = 58$

- 2) In the given frequency distribution, we have

Lower limit of first class = 0

Upper limit of last class = 50

Therefore,

$$\text{Range} = 50 - 0 = 50$$

- 3) First we construct the following cumulative frequency distribution:

Class	f	Cumulative frequencies
0-5	6	6
5-10	8	14
10-15	12	26
15-20	24	50
20-25	36	86
25-30	32	118
30-35	24	142
35-40	8	150
	N=150	

$$Q_1 = L + \frac{\left(\frac{N}{4} - C\right)}{f} \times h$$

15-20 is the Q_1 class

$$= 15 + \frac{\left(\frac{150}{4} - 26\right)}{24} \times 5$$

$$= 15 + \frac{37.5 - 26}{24} \times 5 = 15 + \frac{57.5}{24}$$

$$= 17.40$$

$$Q_3 = L + \frac{\left(\frac{3N}{4} - C\right)}{f} \times h$$

25-30 is the Q_3 class

$$= 25 + \frac{\left(\frac{3 \times 150}{4} - 86\right)}{32} \times 5$$

$$= 25 + \frac{112.5 - 86}{24} \times 5 = 25 + \frac{132.5}{24}$$

$$= 30.52$$

$$\text{Therefore, } QD = \frac{Q_3 - Q_1}{2} = \frac{30.52 - 17.40}{2} = \frac{13.12}{2} = 6.56$$

- 4) We have the following distribution:

x	(x-17)	$ x - \bar{x} $
16	-1	1
24	+7	7
13	-4	4

18	+1	1
15	-2	2
10	-7	7
23	+6	6
$\sum x = 119$		$\sum x - \bar{x} = 28$

Then
$$\bar{x} = \frac{\sum x}{n} = \frac{119}{7} = 17$$

Therefore,
$$MD = \frac{\sum |x - \bar{x}|}{n} = \frac{28}{7} = 4$$

5) First we construct the following frequency distribution:

Class	x	f	xf	(x - $\bar{x}$)	x - $\bar{x}$	f x - $\bar{x}$
0-10	5	5	25	-22	22	110
10-20	15	8	120	-12	12	96
20-30	25	15	375	-2	2	30
30-40	35	16	560	8	8	128
40-50	45	6	270	18	18	108
						$\sum f x - \bar{x} = 472$

Then
$$\bar{x} = \frac{\sum xf}{\sum f} = \frac{1350}{50} = 27$$

and

$$MD = \frac{\sum f|x - \bar{x}|}{\sum f} = \frac{472}{50} = 9.44$$

Check Your Progress 2

1) We have given

$$n = 100, \bar{x} = 16, \sigma = \sqrt{21}$$

$$n_1 = 50, \bar{x}_1 = 20, \sigma_1 = 2, \text{ and } n_2 = 50$$

we have to find

$$\bar{x}_2 = ?, \sigma_2 = ?$$

$$\text{Now, } \bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} = \frac{50 \times 20 + 50 \bar{x}_2}{100}$$

$$16 = \frac{50 \times 20 + 50 \bar{x}_2}{100}$$

$$50 \bar{x}_2 = 16 \times 100 - 50 \times 20$$

$$50 \bar{x}_2 = 1600 - 1000 = 600$$

$$\Rightarrow \bar{x}_2 = 12$$

$$d_1 = \bar{x}_1 - \bar{x} = 20 - 16 = 4 \Rightarrow d_1^2 = 16$$

$$d_2 = \bar{x}_2 - \bar{x} = 12 - 16 = -4 \Rightarrow d_2^2 = 16$$

$$\sigma^2 = \frac{1}{n_1 + n_2} [n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)]$$

$$(\sqrt{21})^2 = \frac{[50(4+16) + 50(\sigma_2^2 + 16)]}{100}$$

$$21 \times 100 - 20 \times 50 = 50\sigma_2^2 + 50 \times 16$$

$$50\sigma_2^2 = 2100 - 1000 - 800$$

$$50\sigma_2^2 = 2100 - 1000 - 800$$

$$50\sigma_2^2 = 300$$

$$\Rightarrow \sigma_2^2 = 6 \Rightarrow \sigma_2 = \sqrt{6}$$

2) First we calculate the following distribution:

Then

x	x ²
10	100
27	729
40	1600
60	3600
33	1089
30	900
10	100
$\sum x = 210$	$\sum x^2 = 8118$

$$\bar{x} = \frac{\sum x}{n} = \frac{210}{7} = 30$$

and therefore,

$$\sigma = \sqrt{\frac{1}{n} \sum x^2 - (\bar{x})^2} = \sqrt{\frac{8118}{7} - 900}$$

$$= \sqrt{1159.7 - 900} = \sqrt{259.7} = 16.115$$

3) Let us take A = 25 and calculate the following frequency distribution:

Class	x	f	d = x-A	f d	f d ²
0-10	5	5	-20	-100	2000
10-20	15	8	-10	-80	800
20-30	25	15	0	0	0
30-40	35	16	10	160	1600
40-50	45	6	20	120	2400
		N = 50		$\sum fd = 100$	$\sum fd^2 = 6800$

Therefore,

$$\begin{aligned}\sigma &= \sqrt{\frac{1}{N} \sum_{i=1}^k f_i d_i^2 - \left(\frac{\sum_{i=1}^k f_i d_i}{N} \right)^2} = \sqrt{\frac{1}{50} (6800) - \left(\frac{100}{50} \right)^2} \\ &= \sqrt{136 - 4} = \sqrt{132} = 11.49\end{aligned}$$

4) We have

$$\begin{aligned}\sigma^2 &= \frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2 \\ &= \frac{1}{n} \times 200 - (4)^2 = 20 - 16 = 4 \\ \Rightarrow \sigma^2 &= 4 \Rightarrow \sigma = 2\end{aligned}$$

Therefore, coefficient of variation

$$\begin{aligned}C.V(X) &= \frac{\sigma}{\bar{x}} \times 100 \\ &= \frac{2}{4} \times 100 = 50\%\end{aligned}$$

5) We have the formulae for coefficient of variation

$$CV = \frac{\sigma}{\bar{x}} \times 100$$

By putting the given values, we have

$$\begin{aligned}\sigma &= \frac{CV \times \bar{x}}{100} = \frac{70.6 \times 16}{100} \\ &= 11.296\end{aligned}$$

Check Your Progress 3

1) Using the formulae, we have

$$\begin{aligned}S_k &= \frac{\text{Mean} - \text{Mode}}{\sigma} \\ 1.28 &= \frac{164 - 100}{\sigma} \\ \sigma &= \frac{64}{1.28} = 50\end{aligned}$$

2) We have given $S_k = 1.2$

$$Q_1 + Q_3 = 200$$

$$Q_2 = 76$$

$$\text{then } S_k = \frac{(Q_3 + Q_1 - 2Q_2)}{(Q_3 - Q_1)}$$

$$1.2 = \frac{(200 - 2 \times 76)}{(Q_3 - Q_1)}$$

$$Q_3 - Q_1 = \frac{48}{1.2} = 40$$

$$Q_3 - Q_1 = 40$$

... (1)

and it is given $Q_1 + Q_3 = 200$

$$Q_1 = 200 - Q_3$$

Therefore, from equation (1)

$$Q_3 - (200 - Q_3) = 40$$

$$2Q_3 = 240$$

$$Q_3 = 120$$

- 3) Let us calculate the mean and median from the given distribution because mode is not well defined.

Class	f	x	CF	$d' = \frac{x-35}{10}$	fd'	fd' ²
0-10	10	5	10	-3	-30	90
10-20	40	15	50	-2	-80	160
20-30	20	25	70	-1	-20	20
30-40	0	35	70	0	0	0
40-50	10	45	80	+1	10	10
50-60	40	55	120	+2	80	160
60-70	16	65	136	+3	48	144
70-80	14	75	150	+4	56	244
					$\sum fd' = 64$	$\sum fd'^2 = 828$

$$\begin{aligned} \text{Median} &= L + \frac{\left(\frac{N}{2} - C\right)}{f} \times h \\ &= 40 + \frac{75 - 70}{10} \times 10 = 45 \end{aligned}$$

$$\begin{aligned} \text{Mean } (\bar{x}) &= A + \frac{\sum_{i=1}^k fd'}{N} \times h \\ &= 35 + \frac{64}{150} \times 10 = 39.27 \end{aligned}$$

$$\begin{aligned} \text{Standard Deviation } (\sigma) &= h \times \sqrt{\frac{\sum fd'^2}{N} - \left(\frac{\sum fd'}{N}\right)^2} \\ &= 10 \times \sqrt{\frac{828}{150} - \left(\frac{64}{150}\right)^2} \\ &= 10 \times \sqrt{5.33} = 23.1 \end{aligned}$$

Therefore, coefficient of skewness:

$$S_k = \frac{3(\text{Mean} - \text{Median})}{\sigma}$$

$$= \frac{3(39.27 - 45)}{23.1} = -0.744$$

3.10 REFERENCES

IGNOU Course Material

1. Post-Graduate Diploma In Applied Statistics (PGDAST), MST 002: Descriptive Statistics, Block 1: Analysis of Quantitative Data, UNIT 2: Measures of Dispersion (for Sections 3.2, 3.3 and related portions in Sections 3.0, 3.1, 3.8, 3.9 and Check Your Progress questions)
2. PGDAST, MST 002: Descriptive Statistics, Block 1: Analysis of Quantitative Data, UNIT 3: Skewness and Kurtosis (for Sections 3.5 to 3.7 and related portions in Sections 3.0, 3.1, 3.8, 3.9 and Check Your Progress questions)
3. Master of Arts in Psychology (MAPC), MPC-006: Statistics in Psychology; Block 3: Normal Distribution, Unit 1: Characteristics of Normal Distribution (for Section 3.4 and related portions in Sections 3.0, 3.1, 3.8, 3.9 and Check Your Progress questions)

Reference Book

4. “Introduction to Probability and Statistics for Engineers and Scientists”, 3rd Edition, Sheldon M. Ross, Elsevier Academic Press, 2004